AD 2. AERODROMES

VOHS AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOHS - RAJIV GANDHI INTERNATIONAL AIRPORT , SHAMSHABAD

VOHS AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	171426N 0782544E North of PTB, ARP Elevation 1975 ft	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	Bearing 210 DEG/20 KM from Husain Sagar	
3	Aerodrome elevation and reference temperature	2030 FT / 39.0 DEG C	
4	Magnetic variation, date of information and annual change	1.25 DEG W (2010) /0.01667 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	GMR Hyderabad International Airport Ltd. Shamshabad, Ranga Reddy- Dist, Telangana, PIN-500108,	
		Telephone:	(91) 40 67394129
		Fax:	(91) 40 67395559
		AFS:	----
		Email:	ceo.ghial@gmrgroup.in
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	1. Air Traffic Services: General Manager (ATM) Airports Authority of India, Shamshabad, Ranga Reddy- Dist, Telangana. PIN-500108 Tel: +91-40-24005201; Fax: +91-40-24008307 Email: gmatmhial@aai.aero 2. Vertivcal Datum : EGM08 All Coordinates are in WGS-84 3. AD Reference Code: 4F	

VOHS AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	Mon to Fri: 0330-1230 UTC (0900-1800 IST) Sat: 0330-0730 UTC (0900-1300 IST) Sun & Holiday: Closed
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24 Prior Arrangement
10	Security	H24
11	De-icing	NIL

12	Remarks	1. The approved hourly RWY traffic handling capacity is as follows: For RWY 27L/09R: MAX NR of ARR and DEP--42 MAX NR of ARR only --34 MAX NR of DEP only --31 2. For RWY 27R/09L when RWY 27L/09R is taken for schedule maintenance: (a) Case 1- When RWY27L/09R is not available for taxiing purpose: MAX NR of ARR and DEP --29 MAX NR of ARR only --18 MAX NR of DEP only --22 (b) Case 2- When RWY 27L/09R is available for taxiing purpose between taxiways A1 and A3 and between taxiways A8 and A10: MAX NR of ARR and DEP --33 MAX NR of ARR only --22 MAX NR of DEP only --26 3. Airlines are required to share with airport Operator (GMR - RGIA, Hyderabad) any reports of reportable incidents which occur on the ground at RGI Airport or during the initial (take off) or final (approach & landing) phases of flight to or from RGI Airport

VOHS AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	By GMR Hyderabad Air Cargo (A Division of GMR Air Cargo and Aerospace Engineering Ltd.)
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	Hydrant and bowzer (12 Dispensers + 07 Refuellers)
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	Available with AIESL & GMRATL with Hanger facility
7	Remarks	NIL

VOHS AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	Novotel Business Brand Hotel with 305 rooms at a distance of 2.75km from PTB and hotels in Hyderabad city.
2	Restaurant(s) at or in the vicinity of aerodrome	Level C: KFC, McDonalds, Coffee Hut, Food street (Multi cuisine Restaurant). Level D: Cafe Coffee Day, Subway, Karachi Bakery, McDonalds, Coffee Hut, Hatti kapi Level F: At visitors' gallery- Coffee Hut, Forecourt : Coffee Café, Check In- Coffee club, Domestic SHA: Monsoon Bar, Barista coffee, Indian Paradise, KFC, Taste of India Domestic bus gates: Coffee club International SHA: Dosa Factory, Indian Paradise, Grab and Fly, Khao Gali, Coffee club Level B & E: Canteen (for Employees of GMR & Airlines)

3	Transportation possibilities	Taxi (Prepaid & Metered) Bus services (TSRTC Town- Volvo, TSRTC Pushpak), Free shuttle bus from Airport to PTC
4	Medical Facilities	24 hrs Medical and Dental facilities provided by Apollo Hospital Enterprises Limited (One full-fledged medical centre housing minimum 17 beds), First Aid counters at Level D, E and F.
5	Bank and post office at or in the vicinity of aerodrome	Banks: Canara Bank at Level C and State Bank of India at Level F ATM: Total 13 Nos. (Level C- 2, Level D- 6 & Level F- 5) Foreign exchange counters: Total 5 Nos. (Level F- 2,Level E- 1 & Level D- 2)
6	Tourist office	TSTDC (Telangana State Tourism Development Corporation at Level D)
7	Remarks	APTDC (Andhra Pradesh Tourism Development Corporation at Level D)

VOHS AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-9 (can be upgraded to CAT- 10 with 01 Hour prior notice)
2	Rescue equipment	Available as per Airport Services Manual Part1 for CAT- 9
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is B744/B773. 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Chief Operating Officer Contact No : +91-40-66602100 Email : Resource.Planning@gmrgroup.in
4	Remarks	NIL

VOHS AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VOHS AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Surface: Rigid Strength: See AD 2.23
2	Designation, width, surface and strength of taxiways	See AD 2.23
3	Location and elevation of altimeter checkpoints	Location All aircraft stands on the passenger terminal apron. Elevation ---
4	Location of VOR checkpoints	At Runway holding positions on TWYs A1, A2, A3 & A9.
5	Position of INS checkpoints	NIL
6	Remarks	Refer Aircraft Parking / Docking for Details

VOHS AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Aircraft Stand ID signs: All stands are provided with stand identification sign and straight lead-in lines Taxiing guidance signage are provided at all intersections and straight sections as recommended by ICAO annex 14. Air Bridges: 44 (Total) -Stand No. 3, 4, 5, 6, 7, 8, 9, 13, 14, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54L, 54R, 55L, 55R, 56L, 56R , 57L, 57R, 58L, 58R, 59L, 59R, 60L, 60R, 61, 62, 63L, 63R, 64, 66, 67, 68 & 69. Visual Docking Guidance Systems (VDGS/A-VDGS):56 (Total) From Stand no. 3, 4, 5, 6, 7, 8, 9, 13, 14, 39, 40, 41,42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54L, 54, 54R, 55L, 55, 55R, 56L, 56, 56R, 57L,57, 57R, 58L, 58, 58R, 59L, 59, 59R, 60L, 60, 60R, 61, 62, 63L, 63, 63R, 64, 65, 66, 67, 68 & 69. For proper response from VDGS/A-VDGS pilots are advised not to exceed taxi speed of 4m per sec while docking
2	Runway and taxiway markings and lights	RWY Markings RWY 09R/27L : Designation /THR / TDZ / CL / Aiming Point / Side strip RWY 09L/27R: Designation / THR / TDZ / CL / Aiming Point / Side strip / Displaced THR RWY Lights RWY 09R: THR/ EDGE/ TDZ/ CL/ END AND CAT-II/III APP RWY 027L: THR/ EDGE/ CL/ END AND CAT-I APP RWY 09L: THR/ EDGE/ THR IDENT LGT/ END AND CAT-I APP RWY 27R: THR/ EDGE/ THR IDENT LGT/ END AND CAT-I APP TWY Marking CL / EDGE / RHP / Enhanced TWY CL / Intermediate Holding PSN Markings. Enhanced taxiway centerline marking provided on taxiways A1, A2, A3, A8, A9, A10, B,B1,B2, B3, B4, B5 and B6 Mandatory instruction marking provided on Taxiways A1,A2,A3,A8,A9,A10,B1,B2, B3, B4,B5 and B6 TWY Lights TWY A1, A2, A3, A8, A9, A10, RET L1, L2, L3, L4, R1, R2, R3, R4, B2, B3, B4, B5, E3, E4, TXL H1, J1, J2, J3, C, C1 with Inset CL LGT & Elevated Edge LGT (for EAST edge H1 & WEST Edge J3). TWY B (between B2 & B6) with inset CL LGT on straight portion and Elevated Edge LGT at junctions. TWY B (west of B6), B1, B2, B6, E, M, M1, K, K1, D, C, TXL H, H1, TXL K3 (North of TXL K3) with Elevated Edge LGT only. Additionally TWY Edge LGT AVBL for RWY 09L / 27R and portion RWY 09R-27L between TWY A1 to A3 and TWY A8 to A10 when used for Taxi Purpose.
3	Stop bars (if any)	Provided on TWY A1, A2, A3, A8, A9 & A10 Stop bar lights provided on TWY B2, B3, B4, B5 and B6 at RWY 09L/27R HLDG PSN. Runway Guard Lights: At every runway holding position point, both sides of taxiway with a pair of elevated guard lights flashing alternate yellow facing the taxiways (A1, A2, A3, A8, A9, A10, B1, B2, B3,B4, B5 & B6)
4	Remarks	Surface Movement Guidance and Control radar has been provided

VOHS AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	POLE	171340.7N 0782656.1E	2034 FT	NIL	MET Antenna Pole
In circling area and at AD	OTHER	171339.8N 0782656.2E	2033 FT	NIL	Electrical Panel Top
In circling area and at AD	NAVAID	171339.1N 0782457.7E	2019 FT	LGTD	GP Antenna 09R Top
In circling area and at AD	NAVAID	171340.8N 0782658.2E	2072 FT	LGTD	GP Antenna 27L Top
In circling area and at AD	BUILDING	171340.8N 0782657.9E	2033 FT	LGTD	GP Hut 27L Top
In circling area and at AD	BUILDING	171339.2N 0782458.1E	1981 FT	LGTD	GP Hut 09R Top
In circling area and at AD	NAVAID	171340.8N 0782700.7E	2043 FT	LGTD	GP 27L Monitor Antenna Top
In circling area and at AD	NAVAID	171339.1N 0782455.1E	1991 FT	LGTD	GP 09R Monitor Antenna Top
In circling area and at AD	TOWER	171405.7N 0782517.9E	2258 FT	LGTD	Lightening Arrester on ATC Tower
In circling area and at AD	BUILDING	171337.8N 0782457.9E	1986 FT	LGTD	Obstacle Light on MET Hut
In circling area and at AD	ANTENNA	171339.2N 0782500.8E	1976 FT	LGTD	RVR Antenna
In circling area and at AD	POLE	171224.4N 0782617.4E	2184 FT	LGTD	TS Electricity Pylon Mast
In circling area and at AD	POLE	171224.5N 0782609.7E	2189 FT	LGTD	TS Electricity Pylon Mast
In circling area and at AD	NAVAID	171349.6N 0782700.2E	2047 FT	LGTD	Windsock Top
In circling area and at AD	NAVAID	171348.0N 0782456.1E	1993 FT	LGTD	Windsock Top
In circling area and at AD	ANTENNA	171339.2N 0782459.8E	1976 FT	LGTD	RVR Antenna
In circling area and at AD	ANTENNA	171340.8N 0782655.1E	2030 FT	LGTD	RVR Antenna
In circling area and at AD	ANTENNA	171340.8N 0782656.1E	2030 FT	LGTD	RVR Antenna
In circling area and at AD	ANTENNA	171340.0N 0782558.0E	1991 FT	LGTD	RVR Antenna (MID)
In circling area and at AD	ANTENNA	171340.6N 0782655.9E	2055 FT	LGTD	Met Antenna 27L
In circling area and at AD	ANTENNA	171339.1N 0782500.6E	2002 FT	LGTD	Met Antenna 09R
In circling area and at AD	ANTENNA	171354.5N 0782508.0E	2011 FT	LGTD	Met Antenna 09L
In circling area and at AD	ANTENNA	171355.9N 0782647.5E	2049 FT	LGTD	Met Antenna 27R
In circling area and at AD	NAVAID	171354.5N 0782503.4E	1994 FT	LGTD	GP 09L Monitor Antenna Top
In circling area and at AD	NAVAID	171354.5N 0782506.0E	2029 FT	LGTD	GP Antenna 09L Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	171354.5N 0782506.2E	1992 FT	LGTD	GP Hut 09L Top
In circling area and at AD	POLE	171339.5N 0782658.5E	2037 FT	LGTD	Arrester On MET Hut

VOHS AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	VOHS International (AMO)
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	AMO Shamshabad (VOHS) 09 HRS & 30 HRS
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend H24. Half hourly
5	Information on how briefing and/or consultation is provided	Personal ConsultationH24. Briefing provided mainly OLBS,Partial self-briefing and Telephone to Forecaster.
6	Types of flight documentation supplied and language(s) used in flight documentation	Charts, abbreviated plain language text. ENGLISH
7	Charts and other information displayed or available for briefing or consultation	SWH (National), SWM (National), SW International, Wind/Temp Charts (FL 050 to FL 530), SIGMET charts from OLBS
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	TELE 040-24005219/24008233,FAX 040- 24008269
9	The air traffic services unit(s) provided with meteorological information	Shamshabad / Begumpet ATC and ACS
10	Additional information, e.g. concerning any limitation of service.	Integrated WX system RWY 09R/27L AVBL for OPR purpose. RVR INSTR RWY 09R/27L available.

VOHS AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
09R	88.91 DEG	4260 x 60 M	890/F/B/W/T Asphalt	THR: 171342.91N 0782446.11E RWY END: 171344.89N 0782710.27E
27L	268.91 DEG	4260 x 60 M	890/F/B/W/T Asphalt	THR: 171344.89N 0782710.27E RWY END: 171342.91N 0782446.11E
09L	88.91 DEG	3707 x 45 M	866/F/A/W/T Asphalt	THR: 171350.35N 0782454.80E RWY END: 171352.08N 0782700.25E
27R	268.91 DEG	3707 x 45 M	866/F/A/W/T Asphalt	THR: 171352.08N 0782700.25E RWY END: 171350.35N 0782454.80E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 1970.0FT TDZ: 1978.0FT	0.95%	NIL	NIL	4380 x 280 M
THR: 2026.0FT TDZ: 2026.0FT	0.95%	NIL	NIL	4380 x 280 M
THR: 1974.0FT TDZ: 1984.0FT	0.92%	NIL	NIL	3827 x 280 M
THR: 2030.0FT TDZ: 2030.0FT	0.92%	NIL	NIL	3827 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
285M x 150M		Existing	1. RUNWAY PCR : First 150M both ends 1201/R/B/W/T, surface type Concrete. 2. Vertical datum: EGM08 3. 0.029% (Min) 0.95% (Max) from RWY09R towards RWY27L Transverse slope 1.50% (throughout)

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
285M x 150M		Existing	1. RUNWAY PCR: First 150M both ends 1201/R/B/W/T, surface type Concrete. 2. Vertical datum: EGM08 3. 0.029% (Min) 0.95% (Max) from RWY09R towards RWY27L Transverse slope 1.50% (throughout)
283M x 90M Displacement 310 M		Existing	Instrument non- precision approach runway 0.084% (min) 0.918% (max) from RWY09L towards RWY27R transverse slope1.5% (throughout)
250M x 90M Displacement 343 M		Existing	Instrument non- precision approach runway 0.084% (min) 0.918% (max) from RWY09L towards RWY27R transverse slope1.5% (throughout)

VOHS AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
09R	4260	4260	4260	4260	
27L	4260	4260	4260	4260	
09L	3707	3707	3707	3707	The displaced portion of RWY 09L and 27R, shall not be available as part of TORA
27R	3707	3707	3707	3707	The displaced portion of RWY 09L and 27R, shall not be available as part of TORA

VOHS AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
09R	CAT III 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.82FT)	900 M
27L	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.82FT)	NIL
09L	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.82FT)	NIL
27R	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (73.82FT)	NIL

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4260 M 15 M LIH White	4260 M 60 M White LIH	Red	NIL	1.Runway centerline lights variable white from the THR to the point 900 m from the runway end; alternate red and variable white from 900 m to 300 m from the runway end; and red from 300 m to the runway end 2.Runway edge lights variable white, Last 600m yellow towards the landing direction
4260 M 15 M LIH White	4260 M 60 M White LIH	Red	NIL	1.Runway centerline lights variable white from the THR to the point 900 m from the runway end; alternate red and variable white from 900 m to 300 m from the runway end; and red from 300 m to the runway end. 2.Runway edge lights variable white, Last 600m yellow towards the landing direction
NIL	3707 M 60 M White LIH	Red	NIL	1.Runway edge lights variable white, Last 600m yellow towards the landing direction 2.Threshold identification light available (flashing white)
NIL	3707 M 60 M White LIH	Red	NIL	1.Runway edge lights variable white, Last 600m yellow towards the landing direction 2.Threshold identification light available (flashing white)

VOHS AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	On top of ATC Tower, 24FPM, white and green, 400,000cd Switched on only during the operation of AD Ground LGTs.
		IBN	NIL

2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Illuminated, White light between Runways 09R/27L & 09L/27R, 145M north of Runway 09R-27L centreline & 80M south of Runway 09L-27R (171348.534N, 0782552.001E) Horizontal and Perpendicular segment are of 4M length with 11 lights each spaced equidistant. Each bulb is of 50W halogen with White lens.
		Anemometer	1. For RWY 09R & 27L one each South of RWY near TDZ coordinates a. Near RWY 09R : 171339.03 N 0782500.69 E b. Near RWY 27L : 171340.08 N 0782656.04 E 2. For RWY 09L & 27R one each North of RWY near TDZ coordinates a. Near RWY 09L : 171354.07 N 0782508.07 E b. Near RWY 27R : 171355.82 N 0782647.08 E
3	Taxiway edge and taxiway centre line lights;	Edge	Elevated blue lights placed at 3m from taxiway edge. The lateral distance between lights are 60m on straight section and on curved portion varies on the radius of the curve.
		Centre Line	Inset Bidirectional Green Light with the gap of 15 M & 30 M on Straight Sections. On Rapid Exit TWY The CL LGT are Unidirectional LGT Showing Green towards the Runway. TWY Edge LGT are Elevated Blue LGT installed at 3 M from the TWY Edge with the gap of 60 M on Straight sections.
4	Secondary power supply including switch-over time;	DG power as standby, 15 sec switchover time. Critical circuits provided with UPS for zero percent switchover time (Threshold/End/Rwy center line/Edge lights)	
5	Remarks	WDI: Two (i) North of RWY 27L Beginning (ii) North of RWY 09R Beginning. Colours: Orange & white bands &illuminated.	

VOHS AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VOHS AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Area bounded by lines joining points 172410N 0783743E then along the clockwise arc of a circle of 15NM radius centred on 171426N 0782544E to 172708N 0781717E then along the clockwise arc of a circle of 10NM radius centred on 172712N 0782745E to point of origin.
2	Vertical limits	5500 FT AMSL
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Shamshabad TWR, English
5	Transition altitude	7000 FT
6	Hours of applicability	H24
7	Remarks	NIL

VOHS AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
OTHER	Hyderabad Control/ Radar	120.950 MHZ	
OTHER	Hyderabad Control/ Radar	128.350 MHZ	
OTHER	Shamshabad Delivery	121.625 MHZ	
TWR	Shamshabad TWR	118.200 MHZ	
TWR	Shamshabad TWR	118.450 MHZ	
ATIS	----	126.475 MHZ	
ALRS	----	121.500 MHZ	
RADAR	Hyderabad APP/Radar	120.250 MHZ	
RADAR	Hyderabad APP/Radar	125.550 MHZ	
SMC	Shamshabad GND	118.200 MHZ	
SMC	Shamshabad GND	121.850 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	H24	200 NM Coverage
	H24	200 NM Coverage
	H24	5 NM Coverage
	H24	(ALTN Frequency) 25 NM coverage
	H24	25 NM coverage
	H24	75NM Coverage
	H24	50 NM Coverage
	H24	75NM Coverage
	H24	75 NM Coverage
	H24	ALTN Frequency 5 NM Coverage
	H24	5 NM Coverage

VOHS AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 09R CAT III	IHBD	108.500 MHz	H24
LOC 09L CAT I	IHYN	108.900 MHz	H24
LOC 27L CAT I	ISAM	110.900 MHz	H24
GP 09R	IHBD	329.900 MHz	H24
GP 09L		329.300 MHz	H24
GP 27L	ISAM	330.800 MHz	H24
DME ILS 09R	IHBD	CH22X	H24
DME ILS 09L	IHYN	CH26X	H24
DME ILS 27L	ISAM	CH46X	H24
DVOR/DME	HIA	113.800 MHz CH85X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
171345.1N 0782725.3E			
171352.5N 0782726.4E			Localizer is restricted in 150 Hz Sector upto 30 degree.
171342.7N 0782431.3E			25 NM Coverage
171339.0N 0782457.7E			Glide angle 3.0 DEG
171354.5N 0782506.0E			1. GP Coverage is restricted at 10NM above 2000 FT HAT (Height above threshold) 2. Glide Angle : 3.0 Deg.
171340.7N 0782658.1E			10 NM coverage
171338.9N 0782457.7E	1990 FT		Vertical Datum: EGM-08
171354.5N 0782506.0E	1994 FT		
171340.7N 0782658.1E	2067 FT		1.25NMCoverage 2. Vertical Datum: EGM-08
171346.7N 0782420.7E	2003 FT		(I) Gravity Model: EGM-08 (II) DME colocated with DVOR.

VOHS AD 2.20 LOCAL AERODROME REGULATIONS

1. RWY 09L/27R available for day and night operations and is a dependent RWY i.e. at a time RWY 09L/27R or RWY 09R/27L can be used for landing or take-off.

2. ATC initiated continuous descent OPR (CDO) in vogue WI Hyderabad TMA, whenever feasible, DIST to touchdown and CLR limit will be specified by ATC for continuous descent OPR. Pilots are required to REP fuel saving for the CDO after LDG on 121.850 MHZ and send a detail debrief to email: vohs.ops@aai.aero

3. Runway Capacity Enhancement Measures:

3.1 Departure Procedure

3.1.1 Taxiing aircraft should maintain a minimum taxiing speed of not less than 15 knots on the straight portion of taxiways and between 8-12 knots during turning manoeuvres during fair weather conditions.

3.1.2 ATC may alter the departure sequence of an aircraft, which is not adhering to the procedure enumerated in sub para 1.1 to optimize the runway utilization.

3.1.3 Pilots shall complete all mandatory pre-departure checks before entering the active runway for departure so that the aircraft is in a position to take off immediately upon receipt of take-off clearance. Pilots not able to comply with this requirement must notify SMC/TWR before reaching the holding point.

3.1.4 When the aircraft is issued with a line-up and take-off clearance at the taxi holding position it shall be in a position to line up and initiate an immediate take-off in one continuous movement, if unable advise ATC.

3.1.5 When the aircraft is issued with a take-off clearance after lining up on the runway it shall commence take-off roll immediately. If the Controller observes a delay in respect of the departing aircraft commencing its take off run after issuance of take-off clearance, the take- off clearance shall be cancelled and the aircraft shall be advised to vacate the runway immediately at the nearest taxiway to make way for the subsequent arrival or departure. Necessary entries in this regard shall be recorded in the Log Book.

3.1.6 All departures requiring, or able intersection departure shall inform Shamshabad ground before requesting taxi.

3.2 Arrival Procedure

3.2.1 The speed control measures published in ENR 1.6 EAIP INDIA shall be followed by flight crew.

3.2.2 Pilots are reminded that rapid exit from the runway enables ATC to apply minimum spacing on final approach that will achieve maximum runway utilisation and will minimise the occurrences of ‘go around’.

3.2.3 Aircraft are expected to vacate the runway via first available RET depending on aircraft performance in order to reduce delays and maximise utilisation. Pilots not able to comply with this requirement/request should notify APP/TWR as early as feasible.

3.2.4 All ATR category aircraft to vacate RWY preferably via RET (rapid exit TWY) L1 when RWY in use 09R and RET (rapid exit TWY) R1 when RWY in use 27L.

3.2.5 All Code C aircraft to vacate RWY preferably via RET (rapid exit TWY) L2 when RWY in use 09R and RET (rapid exit TWY) R2 when RWY in use 27L.

4. Operational Compatibility for Secondary Runway 09L-27R, and Speed Restriction at TWY K-K1:

4.1. Secondary Runway Operations

i. Secondary RWY 09L-27R is compatible for day and night operations to cater unavailaibility of Main RWY 09R-27L due to any exigencyor scheduled/unscheduled maintenance works or other unforeseen events.

ii. SRY RWY 09L/27R shall be compatible for operations up to Code E aircraft.

iii. Arcraft operations shall be restricted to Secondary Runway (09L/27R) and would not be extended to Main Runway

(09R/27L) for taxiing purpose during the Night operations in case portion of RWY 09R-27L between TWY A1-A3 & A8 -A10 not available. Markings for making 180 degree turn is provided abeam TWY A2-B1 and A9-A10 dispalced portion area on SRY RWY 09L27R at the either end portions.

4.2 Following limitations/restrictions will remain in place while activation of secondary Runway.

i. No code F aircraft movement accepted during Secondary Runway operation.

ii. Taxiway B1 upto ATC control location (165 M from SRY RWY Centerline) shall be available for operation and MRO movements shall be accommodated with prior coordination and approval of ATC, ASO and AOCC.

iii. In case of Isolation Bay (Stand No. 100) not available for parking during RWY 09L/27R operations due to operational/ technical difficulties in restoration of main RWY 09R/27L, as an alternate, portion of RWY 09R/27L between A1-A2 or A8-A9 or Portion of TWY B1 beyond RHP shall be used for parking of Aircraft under contigency in coordination with ATC, AOCC,ASG and other concerned department.

4.3. Description of AGL System during activation of Secondary Runway:

Sl. No	AGL Circuits	Secondary Runway 09L-27R	Main Runway 09R-27L	Remarks
1	Runway Threshold lights	ON	OFF	-
2	Runway Edge lights	ON	OFF	-
3	Runway Centreline Lights	Not Applicable	OFF	
4	Runway End lights	ON	OFF	-
5	Approach lights	ON	OFF	-
6	PAPI	ON	OFF	-
7	Beacon Wind cone	ON	-	-
8	Guard Lights	ON	OFF	-
9	TILS	ON	Not Applicable	-
10	Signage's	ON	OFF	-
11	Taxiway Edge and Centreline lights (connecting to Secondary Runway)	ON (for taxiway B2, B3, B4,B5 & B6)	Optional (for TWY A1 to A10)	At the discretion of ATCO or on the request of pilot using portion of RWY 09R-27L between A1-A3 & A8-A10 for Departure or Arrival

Note:

i. The Secondary Runway (09L-27R) Edge lights and Taxiway Edge Lights are interlocked to preclude the possibility of simultaneous operation of both form of lighting.

ii. TWY Edge lights is available for Aircraft guidance to use portion of Main Runway 09R-27L between TWY A1-A3 and TWY A8-A9/10 for Aircraft requiring full length of Secondary Runway for Arrival or Departure.

iii. During daylight and optimum visibility condition ATC can switch off Secondary Runway 09L-27R AGL system except guard lights and Threshold Identification lights.

4.4. Aircraft Taxing Speed Restriction at HOT SPOT- 5 & 6

(HOTSPOT-5: APRON SERVICE ROAD CROSSING TWY- K, TWY- K1 AND SERVICE ROAD CROSSING TWY B3)

HOTSPOT-6: APRON SERVICE ROAD CROSSING TWY- M AND TWY- M1)

To mitigate the Hot-Spot - 5 (TWY K & TWY K1) & Hot-Spot -6 (TWY M & TWY M1), taxing speed restriction of Max 7 knots is imposed. All the pilots are requested to adhere to the same.

5. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM)

5.1 INTRODUCTION

5.1.1 Efficiency of Air Transport System is highly dependent on traffic predictability. A- CDM effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Aircraft Operators, Airport Operator, ATC etc.

5.1.2 A-CDM is a Common Information Sharing Platform introduced to process timely and accurate information for quick and precise decision making. A-CDM takes into consideration the ETAs of arriving aircraft, Target Off Block Times of departures, Runway-in-use and the handling capacity of the runway and determines the arrival and departure sequence. Based on the sequence and the parking position, it calculates the Target Take-Off Times & Target Start up Times of departures and Target In-Block Times for arrivals.

5.2 TWO KEY PARAMETERS OF A-CDM: TOBT and TSAT

5.2.1 TARGET OFF-BLOCK TIME (TOBT)

5.2.1.1 Definition: The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the SMC (GROUND). It is the real time target of ensuring readiness of an aircraft and therefore, more accurate than the static departure time mentioned in the flight plan and hence, an ideal milestone to be used by all airport partners.

5.2.1.2 Communication of TOBT:

i. If no delay is expected, EOBT will be treated as TOBT. Action need not be performed by the Aircraft Operator in the ACDM interface.

ii.For delay within 30 minutes of the EOBT, TOBT shall be updated in the ACDM interface by the aircraft operator. Optionally, EOBT may be revised via AFTN through DLA/CHG message.

iii. For delay more than 30 minutes from the EOBT, EOBT shall be mandatorily revised through a DLA/CHG message. (Option for changing TOBT will not be available in this case).

5.2.1.3 Accuracy: TOBTs must be updated to an accuracy of +/- 5 minutes.

5.2.2 TARGET START UP APPROVAL TIME (TSAT)

5.2.2.1 Definition: The time calculated by A-CDM application taking into account TOBT, CTOT and/or the traffic situation that an aircraft can expect to start-up/push back for departure.

Note: In case there is no congestion, ATC may allow the aircraft actual start up approval (ASAT) even before the TSAT.

5.2.2.2 Calculation of TSAT: TSAT is calculated automatically by A-CDM Application. The TSAT is calculated by taking into account TOBT, priority of aircraft, variable taxi times, and the runway capacity. Any revision of EOBT will trigger the A-CDM application to recalculate a new TSAT for that flight as well as to re-allocate earlier TSAT to next flight who meets the requisite conditions. TSAT is displayed in various ACDM interfaces in the following colour coded information.

•Blue colour: TSAT -15 minutes up to TSAT -5 minutes

•Green colour: TSAT -5 minutes up to TSAT +5 minutes

•Red colour: After TSAT +5 minutes.

5.2.2.3 Generation of TSAT: TSAT is generated by ACDM interface at TOBT – 60 minutes.

5.2.2.4 Communication of TSAT: TSAT will be displayed in various A-CDM interfaces. The AO/GHA shall advise flight crew of TSAT, displayed in the A-CDM interface.

5.2.2.5 Priorities for generation of TSAT:

The following aircraft will be accorded priority by A-CDM application at the time of generating TSATs:

i.VVIP flights

ii.Ambulance aircraft

iii.Aircraft in relief missions

iv.Aircraft returning to stand after push back/taxiing out for any reason.

5.2.2.6 Airlines/GHAs are required to monitor the A-CDM interface regularly to get information on the revised TSAT if any, in respect of their aircraft. It may please be noted that the earlier allocated TSAT may get changed due to the following reasons:

i.Change of runway.

ii.Change in taxi times.

iii.Revised TOBT of the subject aircraft.

iv.Cancellation/ revision in TOBT of other aircraft in the departure sequence.

v.Priority handling.

vi.Unusual Occurrences, etc.

vii.Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay.

viii.Inclement Weather.

ix.Bird activity.*

x.Change of route (SID)*

* [Future provision]

5.2.2.7 Accuracy: The TSAT will remain valid for +/- 5 minutes.

5.3 Co-ordination with the Air Traffic Flow Management Unit (ATFMU) *

5.3.1 A fully automatic data exchange with the ATFMU is established. A-CDM system is receiving ELDT and CTOT from C-ATFM.

5.3.2 The data sent from A-CDM to C-ATFM enables efficient CTOT calculations.

5.4 A-CDM Start-up/Push-back procedures

5.4.1 Pre-Departure Sequence: Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their Gate/Stands. It should not be confused with the Pre- Take-off sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

5.4.2 Start-Up and Push-Back Procedure:

i.The aircraft must be ready for Start-up/Push-back at TOBT. Any revisions in TOBT shall be updated by the aircraft operator at the earliest.

ii. The Pilot will contact Clearance Delivery (CLD) and request for Pre-Departure Clearance between TSAT -15 minutes to TSAT-5 minutes. (Blue Zone)

iii. Pilot shall contact SMC (GROUND) for Start-up/Push-back between TSAT-5 minutes and TSAT+ 5 minutes (Green Zone), subject to change over from CLD.

Note: The system may prepone TSAT up to TOBT

iv. If by TSAT +5 minutes (Red Zone after TSAT+5 minutes), ATC has not received Start-up/Push-back request, the aircraft will lose its position in sequence. In such case:

a. Aircraft Operator shall update the TOBT.

b. A revised TSAT will be issued because of resequencing done based on new TOBT.

v. ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC if there is a delay.

vi. Taxi clearance shall be requested within 5 minutes of Start-up/Push-back approval time.

Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required. Aircraft Operator shall be responsible to obtain new ADC number from IAF-MLU.

vii. The Pre-Departure (Off- block) Sequence will be determined in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request. Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

5.5 INFORMATION DISSEMINATION TO A-CDM STAKEHOLDERS

Relevant information for stakeholders, available with A-CDM system, is disseminated through following means:

i. Access controlled application interface provided to representatives of Airline Operators, Ground Handlers and Airport Operator.

ii. Web interface that can be accessed over Internet.*

iii. Display boards on the airside at certain parking stands, in the viewable area of cockpit crew, where limited information is disseminated.*

* [Future provision]

5.6 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the A-CDM procedures, the guidance material published by Euro control, ICAO and other Organizations has been used].

S. No.	Acronyms	Definition	Explanation
6.1	ELDT	Estimated Landing Time	The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing).
6.2	ALDT	Actual Landing Time	The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = Landing, ACARS = ON).
6.3	EXIT	Estimated Taxi-In Time	The estimated taxi time between landing and in- Block.
6.4	AXIT	Actual Taxi-In Time	AXIT = AIBT – ALDT
6.5	SIBT	Scheduled In- Block Time	The time that an aircraft is scheduled to arrive at its first parking position.
6.6	EIBT	Estimated In- Block Time	The estimated time that an aircraft will arrive in- blocks. (Equivalent to Aircraft Operator's ETA = Estimated Time of Arrival).
6.7	AIBT	Actual In- Block Time	The time that an aircraft arrives in-blocks. (Equivalent to Aircraft Operator's ATA = Actual Time of Arrival, ACARS = IN).
6.8	STTT	Scheduled Turnaround Time	STTT = SOBT – SIBT
6.9	ETTT	Estimated Turnaround Time	The time estimated by the AO/GH on the day of operation to turnaround a flight taking into account the operational constraints.
6.10	MTTT	Minimum Turnaround Time	The minimum turnaround time agreed with an AO/GH for a specified flight or aircraft type.
6.11	ATTT	Actual Turnaround Time	ATTT = AOBT – AIBT
6.12	ASRT	Actual Start Up Request Time	Time the pilot requests start up clearance.
6.13	TSAT	Target Start Up Approval Time	The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an Aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT.
6.14	ASAT	Actual Start Up Approval Time	Time that an aircraft receives its start-up approval.
6.15	SOBT	Scheduled Off- Block Time	The time that an aircraft is scheduled to depart from its parking position.
6.16	EOBT	Estimated Off- Block Time	The estimated time at which the aircraft will start movement associated with departure (ICAO).
6.17	TOBT	Target Off- Block Time	The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND).
6.18	AOBT	Actual Off- Block Time	Time the aircraft pushes back / vacates the parking position. (Equivalent to Aircraft Operator's ATD = Actual Time of Departure & ACARS = OUT)
6.19	ARDT	Actual Ready Time (for Movement)	When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition.
6.20	EXOT	Estimated Taxi-Out Time	The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off.
6.21	AXOT	Actual Taxi- Out Time	AXOT = ATOT – AOBT
6.22	ETOT	Estimated Take Off Time	The estimated take off time taking into account the EOBT plus EXOT.
6.23	CTOT	Calculated Take Off Time	A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne.

6.24	TTOT	Target Take Off Time	The Target Take Off Time taking into account the TOBT/TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between Aircraft.
6.25	ATOT	Actual Take Off Time	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF).

7. PERIODIC CLOSURE OF RUNWAY FOR PREVENTIVE MAINTENANCE:
RWY 09R/27L closed for preventive maintenance every Tuesday between 0630-1000 UTC subject o weather and visibility. RWY restoration time is 30min. During this period Secondary RWY 09L/27R will be available for operations up to Code-E aircraft only.

VOHS AD 2.21 NOISE ABATEMENT PROCEDURES

1. Continuous Climb Operation (CCO) Procedures

a) Objective: To minimize aircraft noise impact on surrounding communities during departure by implementing Continuous Climb Operations (CCO), ensuring a smooth, uninterrupted climb profile with minimal level-off segments.

b) Applicability: All aircraft / turbojet aircraft departing from RGIA between 1830 UTC and 2330 UTC are encouraged to participate in CCO, subject to ATC clearance and operational feasibility.

c) Operational Conditions:
i) CCO will be implemented when radar and SIDs (Standard Instrument Departures) procedures are available by ATC.
ii) CCO must not adversely affect airport capacity or traffic flow.

d) ATC Coordination:
i) Aircraft shall be cleared via published SIDs.
ii) ATC may provide radar vectors for sequencing or separation, ensuring minimal level-off during climb.
iii) Pilots should request CCO clearance during pre-departure clearance or prior to pushback. ATC may suspend or cancel the CCO due to traffic conditions even after CCO is cleared.
iv) Alternate instructions will be issued when CCO is suspended or cancelled.

e) Climb Profile:
i) Aircraft should commence climb immediately after takeoff and maintain continuous climb to cruise altitude as instructed by ATC.
ii) Level-off segments should be avoided unless required for traffic separation or airspace constraints.
iii) Speed and altitude restrictions in SIDs must be followed unless otherwise instructed by ATC.

f) Noise Reduction Measures:
i) Use of reduced thrust takeoff procedures where feasible.
ii) Avoid use of high-power settings near noise-sensitive areas.
iii) Pilots should minimize use of high engine power settings during initial climb.
iv) Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimize the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VOHS AD 2.22 FLIGHT PROCEDURES

I. LOW VISIBILITY PROCEDURE:

1. Definitions and Abbreviations:

1 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

1.2 Low visibility take-off (LVTO). A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m.

1.3 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

1.4 Runway Visual Range: The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

1.5 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a. Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions;

b. Landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude / height (MDA/H) and, if necessary, cloud conditions; and

c. Landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

d. Aerodrome operating minima for Airports shall be established by the airline/aircraft operators on the OCA/OCH or DA/DH and ALS facility published by AAI.

e. The adherence to the minima is the responsibility of pilot in command.

However, ATC is responsible to pass the latest meteorological information and status of aeronautical ground lights and other related facilities at the airport to the pilot in command, including any significant changes

1.6 Safeguarding Procedures: Safeguarding Procedures are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all Safeguarding procedures are complete and airport is configured for Low Visibility Take-offs.

1.7 Precision approach and landing operations: An instrument approach procedure based on navigation systems (ILS, MLS, GLS and SBAS Cat I) designed for 3D instrument approach operations Type A or Baltitude/height (DA/H) appropriate to the type and/or category of the operation.

1.8.1 ILS Category I (Cat I) Operations: A decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m.

1.8.2 ILS Category II (CAT II): A decision height lower than 60 m (200 ft), but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;

1.8.3 ILS Category III (CAT III): A decision height (DH) lower than 30 m (100 ft), or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

1.9 Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

1.10 Visibility: Visibility for aeronautical purposes is the greater of:

a. The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b. The greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background.

Note 1: The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2: The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

1.11 Decision Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Note. — Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.

1.12 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

1.13 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary.

1. 14 The abbreviations used in this document have the following meanings:

AAI: Airports Authority of India

ADC: Aerodrome Control

AGL: Aeronautical Ground Lighting

APOC: Airport Predictive Operations Centre

ARFF: Airport Rescue and Fire Fighting Services

ASMGCS: Advanced Surface Movement Guidance and Control System

ATC: Air Traffic Control

ATIS: Automatic Terminal Information Service

ATM: Air Traffic Management

GHIAL: GMR Hyderabad International Airport Limited

CFT: Crash Fire Tender

DATIS: Datalink Automatic Terminal Information Service

DG: Diesel Generating Set

GP: Glide Path

ILS: Instrument Landing System

ITH: Intermediate Taxi Holding Position

LOC: Localizer

LSA: Localizer Sensitive Area

LVP: Low Visibility Procedure

LVTO: Low Visibility Take-off Procedure

MET: Meteorology

P&E: Projects & Engineering/GEMS

R/T: Radio Telephony

RGIA: Rajiv Gandhi International Airport

RWY: Runway

RVR: Runway Visual Range

SMC: Surface Movement Control

SP: Safeguarding Procedures

SSO : Shift Supervisory Officer

TDZ : Touchdown Zone

TWR SUP: Tower Supervisor

TWY: Taxiway

WSO : Watch Supervisory Officer

UPS : Uninterruptible Power Supply

2. Introduction:

The Low Visibility Procedure (LVP) incorporates safeguarding measures and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

2.1 General:

2.1.1 Rajiv Gandhi International Airport, Shamshabad is equipped with CAT II/III ILS system for RWY 09R & CAT I ILS RWY 27L & Secondary RWY 09L.

2.1.2 ASMGCS is operational. Also, RWY and TWY centreline lights are available which can be used during LVP.

2.1.3 The following equipment shall be serviceable to the required standard to support LVP operations:

a. Aeronautical ground lighting System consisting of Runway (RWY) edge lights, RWY centreline lights, RWY end lights.

b. ILS localizer, Glide path and ILS DME.

c. Taxiway (TWY) edge lights, Stop bar lights, Approach lights.

d. The critical lighting system RWY edge, RWY centre line and RWY end lights are supported by UPS for zero switch over time.

e. Real time RVR reporting System (available for both RWYs). In case RVR equipment is unserviceable, manual RVR will be made available.

f. Switchover time for standby power for ILS and Aeronautical ground lighting system excluding critical lighting system mentioned in para 2.1.3 (b) is a maximum of 15 seconds.

2.2 Reporting RVR:

a. There are three RVR transmissometer located for RWY 09R & 27L at Touch-down Zone (TDZ), MID Point and End Point Zone (END) and two RVR transmissometer located for RWY 09L & 27R at Touch-down Zone (TDZ) and End Point Zone (END). Touch-down zone RVR needs to be reported for CAT I operations, touch-down and mid zone RVR for CAT II operations, touch-down, mid and roll-out zone RVR for CAT III operations. In all cases, touch-down zone will always be controlling, However, if any other RVR is reported and is relevant (operator shall not define relevant depending on runway length/aircraft stopping distance unless approved by FSD, DGCA) it also becomes controlling. The mid zone and roll-out zone can be lower than the touch-down zone provided conditions enumerated in Note 1 below are met.
The following table is to be used for reference:

Type of operation	RVR		
	Touch-down Zone	**Mid zone**	**Roll-out zone**
CAT I	550M	125M	125M
CAT II	300M	125M	125M
CAT III	175M	125M	125M
CAT III (With roll-out Guidance)	75/50M	75/50M	75/50M

Note 1: The use of minimum RVRs in the table above is subject to:
* Operator authorization.
* Aeroplane authorization.
* Flight crew training and qualification; and
* Aerodrome facilities.
Note 2: The use of minimum RVR of 75m or 50m depends on value approved for operators and aeroplanes with roll-out guidance system.
Note 3: The values in bold font are required for the type of operation.

b. When reporting RVR to pilots the TDZ RVR shall always be intimated for the concerned Landing RWY

i. For CAT II Operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

ii. For CAT III Operations - If TDZ RVR is below 300m, MID and END RVR reading shall also be passed. TDZ, MID and END RVR shall be available.

c. The reference RVR value for the implementation and cancellation of LVP shall be the lower of the TDZ & END RVR.

2.3 Safeguarding Procedures (SP):

a. Safeguarding Procedures shall be initiated when the RVR is less than 1200 m or visibility is forecast to deteriorate to 750m or less; and/or the cloud ceiling is 400 ft and forecast to fall to 200 ft or less.

b. Safeguarding Procedures (SP) shall be implemented whenever ATC considers the introduction of Low Visibility Procedure necessary.

c. LVP shall be implemented only when SP is completed. Safeguarding Procedures are necessary procedures to prepare the airport for LVP operations. This includes:

i. Positioning of 1 CFT each at the two predetermined positions (in front of main fire station and in front of satellite fire station on fire access road);

ii. Stopping of all works on the manoeuvring area and Localizer & glide path sensitive area as well as removal of all men and mobile equipment from the said area;

iii. Restrictions on the movement of vehicles on the manoeuvring area and aprons.

iv. Implementation of secondary power supply change over time of a maximum of 1 second for RWY Edge, centre line and End lights (Both circuit) in RVR conditions less than 750 meters. This is achieved by UPS for zero switch over time with 30 mins of power backup in case of commercial powers supply is not restored even after 30 min, the DG shall takeover automatically from UPS.

v. The appropriate Aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may have to be delayed.

vi. The ADM APOC in consultation with Airside operation shall give confirmation that SP is completed to WSO through SMC. This information may be passed on land line or on SMC frequency.

vii. When any required facility defined in Para 1.6 becomes unserviceable during periods of LVP, TWR/SMC shall advise the aircraft and accordingly LVP operations shall be suspended and information to this effect shall be included in ATIS broadcast.

Note: Main RWY 09R-27L Edge lights are served by two Independent interleaved circuits. In case one circuit becomes unserviceable, the RWY light system will meet the stipulated specification for any operation (Precision, Non -Precision & Noninstrument). However, Sec. RWY 09L-27R Edge lights are served by two dependent interleaved circuits at 60 mts and in case one circuit becomes unserviceable, the RWY light system will not meet the stipulated specification for any operation. (Precision, Non -Precision & Non-instrument). Hence, WSO shall take suitable action to inform all concerned regarding suspension of operations till normalcy is restored. The triggers for various activities defined in the LVP procedures are based on RVR/Visibility Data. These triggers are passed to APOC/ARFF by SMC on telephone to initiate the SP.

2.4 Aeronautical Ground Lighting System

a. The Precision Approach CAT II/III lighting is installed for RWY 09R and Precision Approach CAT I lighting system for RWY 27L, RWY 09L and RWY 27R at RGI Airport, Shamshabad for enhanced visual guidance for pilots.

b. All AGL critical circuits RWY edge, RWY centre line and RWY end lights are supported by UPS for zero switch over time with 30 mins of power backup.

c. In case of power failure during low visibility UPS will take over without interruption (seamless transition) without impacting any AGL facility.

d. In case of commercial powers supply is not restored even after 30 min, the DG shall takeover automatically from UPS.

e. When LVP is in force the Aeronautical Ground Lighting shall comply with the following minimum serviceability requirement:

f. The Electrical Power Supply Systems for AGL are so designed that the time interval between failure of the primary source of power and the complete restoration of the services for visual aids associated CAT I operations, shall be in accordance with maximum switch-over times in Table 8-1 of Para 8 of DGCA CAR, Section 4 Series B Part I

g. All AGL critical circuits RWY edge, RWY centre line and RWY end lights are supported by UPS for zero switch over time with 30 minutes of power backup.

h. Stop Bars: Stop bars have been provided as follows

i. On taxiway A1, A2, A3 at holding position for Runway 27L

ii. On taxiway A8, A9, A10 at holding position for Runway 09R

iii) On taxiway B6, B5 at holding position for Runway 09L

iv) On taxiway B2, B3 at holding position for Runway 27R

i. No entry bar: No entry bar lights are installed at RET L1, L2, L3, L4 & R1, R2, R3, R4

j. Taxiway centreline lights provided at Taxiways B2, B3, B4, B5, B6, B, A1, A2, A3, L1, L2, L3, L4, R1, R2 R3, R4, A8, A9, A10, C, C1, E, E3, E4, E1, K, K1, J1, J2, J3, H1, M, M1 and SRY RWY 09L/27R (to be switched on when used as parallel taxiway)

k. Parking stands 50, 51, 52, 53, 54L, 54R, 55L, 55R, 56, 57 are provided with voltage driven CAT III Aircraft Stand Manoeuvring Guidance Lighting system integrated with VDGS and provision of manual override.

l. During LVP procedures in force the Aeronautical Ground Lighting shall comply with the following minimum serviceability requirement in the below table:

Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions
Approach Lights	More than 15% of all lights are unserviceablet	Suspend CAT-I Operations
	Any two consecutive lights or more are unserviceable	
	The inner 450 meters- more than 5 % of all Light	Suspend CAT-II/III Operations
	Beyond 450 meters more than 15% of all Lights	
Runway Edge lights	Any two consecutive lights or more are unserviceable	Suspend CAT-I Operations
	Any one of the two circuits serving the power supply goes unserviceable	
	More than 5% of all Lights	Suspend CAT-II/III Operations
	Two adjacent lamps	

Runway centre line lights	More than 15% of all lights are unserviceable	Suspend CAT I Operations
	Any two consecutive lights or more are unserviceable	
	More than 5% of all Lights are unserviceable	Suspend CAT-II/III Operations
	Any two consecutive lights or more Operations are unserviceable	
Touchdown Zone lights	More than 10% of all Lights	Suspend CAT-II/III Operations
	Two lamps in Barrette.	
Threshold lights	More than 5% of all Lights	Suspend CAT-II/III Operations
	Two adjacent lamps	
Runway End lights	More than 25% of all Lights	Suspend CAT-I/II/III Operations
	Two adjacent lamps	
Taxiway Centre line lights	Two adjacent lamps (RVR conditions less than a value of 350 m)	Suspend CAT-II/III Operations
Standby DG Backup	Generators in substation	Suspend CAT-I/II/III Operations for affected runway

m. Suspension of CAT-II/III Operations: The following equipment shall be serviceable to the required standard to support CAT II/III
operations: -
i) ILS localizer Transmitters
ii) ILS glide path Transmitter
iii) ILS DME.
iv) Airfield ground lighting system (AGL) according to the requirement of mentioned in the table below
v) Instrumented RVR system: TDZ & MID.
vi) Standby power supply for ILS.
vii) UPS/ Standby DG Set power for aeronautical ground lighting system.
viii) Surface Movement Radar SMR for CAT II (for RVR below 350M).

Note 1: DGCA CAR Section 4 Series B Part 1, Para 9.8.7. -- Surface movement radar for the manoeuvring area shall be provided
at an aerodrome intended for use in runway visual range conditions less than a value of 350 m.
Note 2: When any equipment and facilities listed above become unserviceable during periods of LVP, the concerned ATC unit shall advise all aircraft and accordingly suspend CAT II Operations. Information to this effect shall be included in ATIS broadcast.
NOTE 3: Under CAT II/III operations (As applicable), for stop bar:
i) No more than two lights will remain unserviceable; and
ii) Two adjacent lights will not remain unserviceable unless the light spacing is significantly less than that specified.
(Ref Para 10.5.8, DGCA- CAR Section 4 Series B Part I).
NOTE 4: For operations in RVR less than a value of 350 m, no two adjacent taxiway Centre line lights be unserviceable.
NO TE 5: Un-serviceability of any of the following facilities does not affect CATII/III Operations:
i) PAPI
ii) Taxiway edge lights on curves.
iii) Taxiway clearance lights wherever provided.
NOTE 6: ILS sensitive and critical areas has been established for Runway 09R for CAT II/III (As applicable) & RWY 27L, RWY 09L for CAT-I operations. Refer AD 2.24 for ILS critical and sensitive areas chart. Protection of these areas during CAT-I/II/ III Operations will be ensured as per Standard Operating Procedures.

2.5 Selection of Runway in Use:

a. As both, RWY 27L (CAT I) and RWY 09R (CAT II/III), and secondary RWY 09L (CAT I) are equipped with RVR instrument and RVR data is available to working controllers, Tower shall determine Runway in Use depending on prevailing wind conditions during LVP.

b. In case of non-availability of Real time RVR, Manual RVR shall be obtained for the selected runway in use, prior to each departure.

c. In case one of the RVR instrument is unserviceable, preference may be given to the RWY for which instrument RVR data is available during NIL wind condition.

3. Implementation of Safeguarding Procedures (SP) and VISIBILITY PROCEDRUES (LVP):

3.1 Safeguarding Procedures shall be initiated:

a. The RVR is less than 1200m or visibility is forecast to deteriorate to 750m or less; and/or

b. The cloud ceiling is 400ft and forecast to fall to 200ft or less.

3.2 WSO/Tower Supervisor will inform APOC for co-coordinating with all the concerned agencies for implementation of Low Visibility Procedures.

3.3 Actions to be taken for initiation of Safeguarding Procedures by WSO:

a. When meteorological conditions meet the criteria for initiation of SP as stipulated in Para 3.1 WSO/Tower Supervisor will inform APOC, the Communication/Technical Supervisor and the Aerodrome Tower Controller.

b.When all the concerned agencies have completed their necessary actions they shall report to WSO (ATC) /Tower Supervisor that their Safeguarding Procedure (SP) is completed and the airport is safeguarded for LVP operations.

3.4 When Met reported visibility reduces to 2500 Meters:

a. Surface Movement Control in consultation with Aerodrome Control Tower shall inform ARFF and AOCC as per visibility /Weather standby criteria.

b. APOC/Air Side Operation shall stop all work and withdraw all personnel working on the Runway strip and the manoeuvring area and inform SMC.

c. Vehicles equipped with VEGA transponder only will be allowed on the manoeuvring area.

d. SMC shall inform ARFF/APOC whenever visibility improves above 2500 Meters to enable ARFF/ APOC/Air Side Operation to resume normal operations.

e. ARFF shall implement procedures applicable for visibility stand and maintain stand by at station with readiness of CFT's to respond any eventuality.

3.5 When visibility reduces to 1500 Meters:

a. SMC shall inform APOC/Air Side Operation when visibility reduces to 1500 Meters.

b. APOC/Air Side Operation shall coordinate with Duty Manager, Technical Services to ensure that UPS and DG set are performing normally.

c. Duty Manager/Air Side Operation shall ensure that Aeronautical light inspection is initiated and report status of Aeronautical lighting system on completion of the inspection to SMC. SMC shall permit the Aeronautical Light Inspection and accord priority for the same.

d. Met department shall provide instrument RVR data for RWY 09R, RWY 27L & Sec. RWY 09L to the ATC. In case of unserviceability of the RVR instrument, the same shall be intimated to ATC and manual RVR shall be provided. For this purpose, Air Side Operation shall provide Leader vehicle, whenever required and inform Security Operations Control Centre (SOCC) to give priority to the Met team to avoid any delay because of security checks near entry gate.

e. ARFF shall position one CFT each at the two predetermined position after obtaining permission from SMC on R/T when visibility reduced to 1000m or RVR less than 1200m (.)

3.6 When RVR reduces to 750 Meters:

a. SMC shall inform WSO/Tower Supervisor/APOC whenever RVR reduces to 750 Meters;

b. APOC/Air Side Operation shall ensure that the towing of aircraft is done under escort of Leader vehicles and inform ATC of any deviation

c. SMC and APOC/Air Side Operation shall not permit any ground run that needs to be performed on the manoeuvring area except idle power run on the stands;

d. APOC/Air Side Operation and SMC shall ensure that "Follow Me" services are provided to pilots on request;

e. APOC Manager shall inform all users of the Apron of the imposition/withdrawal of low visibility procedures;

f. SMC shall inform APOC whenever RVR improves above 750 meters;

g. APOC/Air Side Operation shall confirm completion of safe guarding procedures to Aerodrome Control Tower.

h. ADC shall operate Stop Bars whenever RVR reduces to 750 meters or below.

3.7 **Low Visibility Procedures (LVP)**

3.7.1 Criteria for implementation of Low Visibility Procedures:

Low Visibility Procedures shall be implemented when –

a) Either, TDZ, MID or END RVR is less than 750 m; and/or Cloud ceiling is less than200 ft; and

b) Safeguarding Procedures (SP) have been completed and the airport is safeguarded.

(Note: Though LVP is implemented when RVR is less than 750m, ILS CAT I operation will continue till TDZ RVR is not less than 550m)

3.7.2 Actions to be taken upon Implementation of Low Visibility Procedures

3.7.2.1 When meteorological conditions meet the criteria for LVP as stipulated in Para 3.7.1, WSO /Tower Supervisor shall implement Low Visibility Procedure and inform:

a) Aerodrome Tower controller.

b) Approach Radar Controller

c) Communication/Technical Shift Supervisory Officer (SSO)

d) Duty Met. Officer

WSO will also ensure that "LOW VISIBILITY PROCEDURE IN FORCE" is included in ATIS Broadcast.

3.7.2.2 Upon implementation of LVP, Tower Supervisor will inform:

a) Fire station

b) APOC

c) include "LOW VISIBILITY PROCEDURE IN FORCE" in ATIS Broadcast.

3.8 Cancellation of Safeguarding Procedures (SP) and Low Visibility Procedures (LVP)

3.8.1 WSO/Tower Supervisor may terminate LVP when –

a) Meteorological conditions improve and TDZ, MID and END RVR are 750 m or more and the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways.

b) Facilities and equipment (listed in Para 2.1.3) necessary for CAT I/CAT II/CAT III (As applicable) operations for RWY 09R are degraded and/or the prevailing conditions are considered unsafe for such operations.

3.8.2 WSO/ Tower Supervisor should consult Meteorological Office for forecast before cancelling SP and LVP.

3.8.3 On cancelling LVP, Aerodrome Control shall include it in the subsequent two ATIS broadcasts that "LOW VISIBILITY PROCEDURES ARE CANCELLED". Aerodrome Tower Controller will inform all the concerned agencies.

3.8.4 If SP are completed and LVP are not subsequently implemented and meteorological conditions improve and the visibility/RVR is more than 1200 m and the cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP.

4. <u>Low Visibility procedure for Departures:</u>

The low visibility procedure for departures shall be implemented when RVR reduces below 750 Meters and/or the cloud ceiling is less than 200ft.

Note: Though LVP is implemented when RVR is less than 750 m, ILS CAT I operations for Main RWY 27L & Sec. RWY 09L will continue till TDZ RVR is not less than 550 m and ILS CAT-III operations for Main RWY 09R shall continue till TDZ RVR is not less than 300 m.

a. The safe guarding procedure shall be completed prior to commencement of Low Visibility Procedure Operations.

b. SMC shall inform WSO/ Tower Supervisor/ Duty Manager APOC/Air Side Operation whenever RVR reduces below 750 Meters who in turn shall inform Duty Manager, Technical Services to check the UPS and DG serviceability and inform DM ASO.

c. When SMC gets the information from APOC about the UPS and DG serviceability report, SMC shall inform WSO/ Tower Supervisor/ Duty Manager AOCC/Air Side operations to start LVP operations.

d. Commercial power supply with UPS back up shall be the primary source when LVP becomes operational

e. In case of power failure during low visibility, UPS will take over without interruption (seamless transition) without impacting any AGL facility.

f. In case of commercial powers supply is not restored even after 30 min, the DG shall takeover automatically from UPS.

g. The number of the vehicles shall be restricted to bare minimum and records of all operating on the manoeuvring area shall be maintained by APOC. These records can be in the form of electronic records or audio records.

h. SMC & APOC/Air Side Operation shall permit towing of only one aircraft at a time on the Apron under escort of Leader vehicles.

i. The following may be included in DATIS "**LOW VISIBILITY PROCEDURES IN FORCE**"

j. TWR shall permit departures only from the beginning of the Runway in use.

k. SMC shall inform WSO/Tower Supervisor/ APOC/Air Side Operation whenever RVR improves above 750 Meters. Duty Manager, APOC/Air Side Operation shall inform Duty Manager, P&E Services when RVR improves above 750 Meters and trend forecast suggests upward improvement to enable transfer of power supply from DG set to Main power.

l. If instrument RVR is unserviceable during the above period, manual RVR readings shall be obtained by Meteorological Personnel prior to each departure. For this purpose, Air Side Operation shall provide necessary vehicle whenever required and inform Security Operations Control Centre (SOCC) to give priority to the Met team to avoid any delay because of security checks near entry gate.

This may also include requirement of female CISF frisking staff to facilitate quick entry of female IMD staff through Airside entry Gate-1.

Note- when RVR reaches 400 mts operations will be stopped at Secondary RWY 09L-27R.

4.1 RVR reduces below 200 Meters:

a. All operations will be stopped.

b. An aircraft may be allowed to depart below 200 metres, if within its minima.

c. Tower Controller shall make log entry of such aircraft and relevant signal to be sent to all concerned as per internal procedure.

4.2 RVR improves above 200 Meters:

a. SMC shall inform APOC/Air Side Operation whenever RVR improves above 200 Meters.

b. The respective agencies follow the procedures mentioned in para 4 and para 3 respectively as visibility improves.

4.3 **Low Visibility Procedure Taxi Route**

4.3.1 Once LVP is declared, AGL team will ensure that the lead-in lights of all CAT II/III (As applicable) compatible stands are switched ON.

4.3.2

a. During CAT II/III conditions (As applicable) "FOLLOW ME" service will be provided to arriving /departing aircraft on request. "FOLLOW ME" service can be put into use up to a minimum visibility of 50m.

b. If any immediate portion of exit taxiway is not available, ATC will guide the aircraft to the nearest exit through ASMGCS.

4.3.3 Person providing "FOLLOW ME" service shall be trained and fully familiar with the taxi routes intersections and other manoeuvring area/apron/parking stands.

4.3.4 During CAT II/CAT III ILS operations (As applicable), the preferred taxi routing as per runway in use is provided below:

ARRIVAL - RWY 09R
RWY09R --> TWY L4/A3/A2/A1 --> TAXI VIA RWY 09L/27R --> TWY B4/E3/E OR TWY B5/E4/E --> PARKING STAND 50, 51, 52, 53, 54L, 54R, 55L, 55R, 56, 57
DEPARTURE - RWY 09R
PARKING STAND 50, 51, 52, 53, 54L, 54R, 55L, 55R, 56, 57 --> TWY E/M1/B6 --> TAXI VIA RWY 09L/27R -->TWY A9/A10

4.4 SUMMARY OF THE LOW VISIBILITY PROCEDURES

4.4.1 Subject to completion of Safeguarding Procedures, LVP comes into operation:

a. When either TDZ, MID or END RVR below 750m, and/or

b. Cloud ceiling below 200ft

4.4.2 Vehicular movement

a. Vehicles shall not be cleared to cross the runway, once an inbound aircraft is 8NM from touchdown.

b. Vehicular movement on the Manoeuvring area shall be restricted to essential vehicles.

c. During CAT II /CAT III operations (As applicable) only vehicles equipped with transponder (Vehicle Locator) shall operate in Manoeuvring area. However, other vehicles crossing taxiways on service road shall be regulated by official of Apron Control by deploying man power.

d. Vehicles shall not be held at any point closer to the runway than the CAT II/ CAT III holding point/stop-bar (As applicable).

4.4.3 Aircraft Movement

a. Aircraft shall not be held at any point closer to the runway than the CAT II/CAT III holding point/stop-bar (As applicable).

b. Aircraft which have landed are not to be instructed to hold on a runway turnoff.

5. Summary of actions by various Units.

5.1 Before commencement of winter season, a meeting will be held by Chief Operating Officer, Hyderabad International Airport Private Ltd. and Regional Executive Director (Southern Region)/ GM (ATM), RGI Airport, AAI in the month of November every year to inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

5.2 All the agencies shall ensure that staff and drivers are suitably trained during CAT II/III operations (As applicable).

5.3 Action by Watch Supervisory Officer (WSO)/Tower Supervisor, AAI

5.3.1 Implementing Safeguarding Procedures - When RVR is less than 1200m and visibility is forecast to deteriorate 750m or below and/or the cloud ceiling is 400ft and is forecast to fall to 200ft or less, tower supervisor in coordination with WSO will inform -

a. APOC,

b. Communication/ Technical supervisor (SSO), AAI

5.3.2 Implementing LVP

5.3.2.1 Tower supervisor in coordination with WSO shall implement Low Visibility Procedures when either:

a. TDZ, MID or END RVR is less than 750m; and/or

b. Cloud ceiling is less than 200ft. and

c. Safeguarding Procedures (SP) have been completed and the airport is safeguarded.

6. Actions by other Agencies (Airlines, Refuelling companies, catering agencies, etc.)

a. All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents

b. All the vehicles must have their Vega/obstruction lights "ON" during Low Visibility Procedures operations.

c. Follow all instructions/sign boards provided for vehicular movement area/service roads.

5.3.2.2 Tower supervisor in coordination with WSO shall inform Aerodrome Tower Controller and Approach/Radar Controller, Duty Met Officer and Communication/Technical Shift Supervisor Officer (SSO) and WSO to ensure that "LOW VISIBILITY PROCEDURE IN FORCE" is included in ATIS broadcast.

5.3.3 WSO/Tower supervisor may terminate LVP when;

a. Meteorological conditions improve and TDZ, MID and END RVR are 750 m or more and the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways.

b. Facilities, equipment and services necessary for CAT II/III operations (As applicable) are degraded and/or the prevailing conditions are considered unsafe for such operations.

5.3.3.1 WSO/Tower Supervisor will intimate Aerodrome Tower Controller, Approach/Radar Controller and Communication Technical Shift Supervisory Officer (SSO) regarding the Termination of LVP operation.

5.4 Action by Tower Controller:

5.4.1 On being notified by WSO that ILS CAT II/CAT III Low Visibility Procedures (As applicable) are to commence, the Aerodrome Tower Controller will:

a. Inform Aerodrome Rescue and Fire Fighting Services and APOC,

b. Check ILS status

c. Check lighting is correctly selected and operating properly

d. Check transmissometer displays

e. Check ASMGCS (for operations below RVR 350 m)

5.4.2 After the commencement of ILS CAT II/III operations (As applicable), the Aerodrome Tower Controller shall:

Check ATIS broadcast and include the message that "ILS CAT II/CAT III Low Visibility Procedures in Force (As applicable)".

a. Give landing clearance to aircraft not later than 2NM from touchdown.

b. Inform changes in RVR readings to the landing aircraft.

c. Give an unimpeded taxi route to arriving aircraft to allow it to clear the Localizer Sensitive Area expeditiously.

d. Inform pilots of all failures of ILS, lighting system, transmissometers relevant to ILS CAT II/ CAT III Low Visibility Operations (As applicable).

e. Initiate emergency action if aircraft on CAT II/III ILS (As applicable) is not seen (on radar display or otherwise) or is not in radio contact, as expected.

5.4.2.1 Record of the above actions with time be maintained and signed by the officer taking action.

5.5 Action by Approach/Radar Controller

5.5.1 On being advised by WSO that ILS CAT II/CAT III Low Visibility Procedures (As applicable) are in force, the Approach/Radar Controller shall:-

a. Inform the arriving aircraft "ILS CAT II/CAT III Low Visibility Procedures in Force (As applicable)".

(Note: Ensure that Pilot acknowledges of being cleared for ILS CAT II/CAT III approach (As applicable).

b. Inform TDZ RVR to arriving aircraft and in addition:

i. For CAT II operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

ii. For CAT III operations – If TDZ is below 300m, then MID and End RVR readings shall also be passed.

Note: After an aircraft is 8NM from Touch Down, RVR observations need not be passed unless there is changes in RVR values.

c. Vector the aircraft to intercept the localizer not less than 10NM from touchdown.

d. Not subject an aircraft to any speed control when within 20NM from touchdown.

e. Issue landing clearance to arriving aircraft not later than 2NM from touchdown.

5.6 Action by Surface Movement Controller

During the period the Low Visibility Procedures are effective the Surface Movement Controller shall:

a. Monitor all surface movement of aircraft and vehicles on the Manoeuvring area.

b. Inform all taxiing aircraft of the preceding taxiing or holding aircraft.

c. Hand over only one aircraft at a time to Tower Controller.

d. Permit only the vehicles equipped with transponder in the manoeuvring area during CAT II/III operations (As applicable). However, other vehicles crossing taxiway crossings on service road shall be regulated by official of Airside Operations, HIAL by deploying manpower.

5.7 Action by Communication/Technical/Electronics Shift Supervisory Officer (SSO)

5.7.1 On receipt of "Outlook for LVP" from the WSO, the Communication/Technical/Shift Supervisory Officer [SSO] will inform the Duty Officer, Equipment Room and have the ILS equipment and its status indicators in ATC units checked up.

He will inform WSO of any unserviceability in the equipment which is likely to affect ILS CAT II/CAT III operation (As applicable).

5.7.2 On receipt of "Advisory Message" from WSO that LVP are to be made effective SSO will maintain continuous watch on the performance of ILS equipment and will inform WSO of any un-serviceability which may affect ILS CAT II/CAT III operation (As applicable).

5.8 Actions by Airside operations & Airport Operations Control Centre (ASO/APOC).

a. APOC/Airside Operation shall stop all work and withdraw all personnel working on the Runway strip and the manoeuvring area and inform SMC.

b. APOC/Air Side Operation shall coordinate with Duty Manager, P&E Services to ensure that UPS and DG set are performing normally.

c. APOC/Air Side Operation shall ensure that Aeronautical light inspection is initiated and report status of Aeronautical lighting system on completion of the inspection to SMC. SMC shall permit the Aeronautical Light Inspection and accord priority for the same.

d. Vehicles equipped with VEGA transponder only will be allowed on the manoeuvring area.

e. APOC/Air Side Operation shall confirm completion of safeguarding procedures to Aerodrome Control Tower.

f. APOC/Air Side Operation shall ensure that the towing of aircraft is done under escort of Leader vehicles and inform ATC of any deviation.

g. APOC/Air Side Operation and SMC shall ensure that "Follow Me" services are provided to pilots on request.

h. APOC Manager shall inform all users of the Apron of the imposition/withdrawal of low visibility procedures.

i. The number of the vehicles shall be restricted to bare minimum and records of all operating on the manoeuvring area shall be maintained by APOC. These records can be in the form of electronic records or audio records.

j. If instrument RVR is unserviceable during the above period, manual RVR readings shall be obtained by Meteorological Personnel prior to each departure. For this purpose, Air Side Operation shall provide necessary vehicle whenever required and inform Security Operations Control Centre (SOCC) to give priority to the Met team to avoid any delay because of security check near entry gate.

7. Termination of Low Visibility Procedures:

a. When metrological condition improves to 750M or more, cloud ceiling is 200 feet or Higher and trend is for improvement, WSO/Tower Supervisor would terminate operations of LVP. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions for the purpose of termination of LVP operations.

b. The WSO/Tower Supervisor will intimate the followings regarding termination of LVP operations.

i. APOC/Airside Operations

ii. Approach/Radar Controller

iii. Communication/Technical Shift Supervisory Officer (SSO)

c. On cancelling LVP, Aerodrome Control shall include it in the subsequent two ATIS broadcasts that "LOW VISIBILITY PROCEDURES ARE CANCELLED".

d. If SP are implemented and LVP are not subsequently initiated and meteorological conditions improve and the visibility /RVR is more than 1200m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP.

8.Applicability

LVTO is applicable to Airport Operator, ATS provider, and to those airline operators having specific DGCA approval as per Civil Aviation Requirements issued by DGCA India under Section 8, Series 'C', Part I. Pilots are required to report their approved take off minima (in metres) to ATC on Ground Frequency (121.85 MHz).

II. TRANSPONDER OPERATING PROCEDURES ON GROUND

1. Introduction:

Advanced Surface Movement Guidance and Control System (A-SMGCS) using Mode-S Multi-lateration has been commissioned Airport. The Aircraft Transponder Operating Procedures, particularly in the movement area of the airport(s), where A-SMGCS has been commissioned, is as given below:

2. DEPARTURE

i. **At the Gate/Stand**:

Select STBY

Enter the discrete SSR code received from Clearance Delivery/Surface Movement Control. Enter the three letter ICAO designator followed by the flight identification number (e.g. AIC748) through the FMS or the Transponder control panel, depending on the avionics.

ii. **On requesting Pushback/Taxi (whichever is earlier):**

Select Transponder or equivalent and AUTO if available

This action will enable the aircraft ID, used as the Call sign by ATC, to be displayed on the surveillance display of ATC. ATC can verify the data and use it for necessary identification.

iii. **When Lining up:**

Select TCAS

Select TCAS only after receiving the clearance to line up, to ensure that the performance of systems based on SSR frequencies (including airborne TCAS units, SSR and A-SMGCS) is not compromised.

3.ARRIVAL

a. **When on the Runway:**

Keep TCAS selected

b. **After vacating the Runway:**

Select Transponder or equivalent and AUTO if available

There is a need that the Transponder remains able to exchange data with the A-SMGCS system. However to ensure that the performance of systems based on SSR frequencies (including airborne TCAS Unit, SSR & A-SMGCS) is not compromised, TCAS shall be deselected when vacating the Runway.

4.Parked on Stand:

Select STBY

The Transponder will not reply to interrogation. The discrete SSR Code given to that particular flight can now be recycled for other flights.

Note: When on ground the aircraft must squawk Mode C, in order to provide the altitude information to the surveillance system, and thereby prevent

a. Clutter on Terminal Approach Radar Display (and)

b. False automatic detection of departure for aircraft still on ground.

III Following measures measures are being implemented for reducing RTF congestion and enhancing Operational efficiency

a. Departing aircraft passing 1000ft to change over to HYD APP (freq 120.25Mhz)unless otherwise specified by ATC tower

b. Departing aircraft on first contact with approach shall Transmit only the callsign, altitude passing and assigned SID in all other cases of first contact with an approach unit, aircraft shall transmit only it's callsign.

c. ACARS equipped aircraft to obtain pre departure clearance through datalink whenever clearance delivery is active.

d. Aircraft shall comply with speed control whenever imposed by ATC.

e. Aircraft to vacate RWY via rapid exit TWY as assigned by tower on final approach or inform inability to ATC.

f. Arriving aircraft on vacating RWY to contact ground on freq 121.85 Mhz unless specifically instructed by tower.

h. Arriving aircraft on reaching the allotted stand need not contact ATC for informing the same.

i. Arriving and departing aircraft to minimize RWY occupancy time for enhancing RWY capacity.

j. Aircraft not to demand time to touch down and direct routings. ATC whenever practicable issue the same

VOHS AD 2.23 ADDITIONAL INFORMATION

I. PARKING STANDS

Stand No.	PCR/ Surface	Co-ordinates	Remarks
3	869/R/A/W/T Concrete	171408.82N 0782555.24E	Deep Pushback on to taxilane K3 up to Nose wheel stop position marking and pull forward on taxilane K2 facing North East abeam stand 5 and then engine startup approved. For Towing guidance white dashed line provided. **Suitability:** Code C **Elevation (m):** 607.000
4	869/R/A/W/T/ Concrete	171408.94N 0782556.71E	Pushback on TXL K2 facing East, pull abeam stand 4 and then engine start-up approved. Pushback guidance line provided with white dashed line. **Suitability:** Code C **Elevation (m):** 607.000
5	906/R/A/W/T/ Concrete	171409.42N 0782558.48E	Pushback on TXL K2 facing North East, pull abeam stand 5 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.000

6	906/R/A/W/T/ Concrete	171409.96N 0782559.72E	Pushback on TXL K2 facing North East, pull abeam stand 6 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.000
7	906/R/A/W/T/ Concrete	171410.54N 0782600.96E	Pushback on TXL K2 facing North East, pull abeam stand 7 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.000
8	906/R/A/W/T/ Concrete	171411.13N 0782602.48E	Pushback on TXL K2 facing East, pull abeam stand 8 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.000
9	906/R/A/W/T/ Concrete	171411.15N 0782603.96E	Pushback on TXL K2 facing East, pull abeam stand 9 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.000
10	906/R/A/W/T/ Concrete	171411.11N 0782605.43E	Pushback on TXL K2 facing East, pull abeam stand 10 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.700
11	906/R/A/W/T/ Concrete	171411.13N 0782606.91E	Pushback on TXL K2 facing East, pull abeam stand 11 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.800
12	906/R/A/W/T/ Concrete	171411.15N 0782608.28E	Pushback on TXL K2 facing East, pull abeam stand 12 and then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.900
13	906/R/A/W/T/ Concrete	171409.63N 0782607.47E	Pushback on TXL K1 facing North / South, abeam stand 14 then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.800
14	906/R/A/W/T/ Concrete	171408.21N 0782607.50E	Pushback on TXL K1 facing North / South, pull abeam stand 14 then engine start-up approved. **Suitability:** Code C **Elevation (m):** 607.900
15	869/R/A/W/T/ Concrete	171409.62 N 0782618.3 E	Pushback facing East or West on TWY D, Pull abeam stand 15 and then engine startup approved **Suitability:** Code C **Elevation**: 610.000 M
16	869/R/A/W/T/ Concrete	171409.63N 0782619.68E	Pushback facing East or West on TWY D, Pull abeam stand 16 and then engine startup approved **Suitability:** Code C **Elevation:** 610.470 M
17	869/R/A/W/T/ Concrete	171409.65N 0782621.05E	Pushback facing East or West on TWY D, Pull abeam stand 17 and then engine startup approved **Suitability:** Code C **Elevation:** 610.860 M
18	869/R/A/W/T/ Concrete	171409.67N 782622.42E	Pushback facing East or West on TWY D, Pull abeam stand 18 and then engine startup approved **Suitability:** Code C **Elevation:** 611.270 M
19	869/R/A/W/T/ Concrete	171409.70 N 0782624.52 E	Pushback facing East or West on TWY D, Pull abeam stand 19 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 611.960 M
20	869/R/A/W/T/ Concrete	171409.24 N 0782625.62 E	Pushback facing East or West on TWY D, Pull abeam stand 20 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 612.320 M

21	906/R/B/W/T/ Concrete	171409.26 N 0782626.99 E	Pushback facing East or West on TWY D, Pull abeam stand 21 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 612.760 M
22	869/R/A/W/T/ Concrete	171409.28 N 0782628.76 E	Pushback facing East or West on TWY D, Pull abeam stand 22 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 613.230 M
23	869/R/A/W/T/ Concrete	171409.30 N 0782630.16 E	Pushback facing East or West on TWY D, Pull abeam stand 23 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 613.640 M
24	869/R/A/W/T/ Concrete	171409.31 N 0782631.53 E	Pushback facing East or West on TWY D, Pull abeam stand 24 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 614.000 M
25	869/R/A/W/T/ Concrete	171409.33 N 0782632.90 E	Pushback facing East or West on TWY D, Pull abeam stand 25 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 614.450 M
26A	869/R/A/W/T/ Concrete	171410.38 N 0782634.39 E	Pushback facing North or South on TWY H, Pull abeam stand 26A and then engine start-up approved **Suitability:** Code B **Elevation:** 614.510 M
26B	869/R/A/W/T/ Concrete	171409.35 N 0782634.50 E	Pushback facing North or South on TWY H, Pull abeam stand 26B and then engine start-up approved **Suitability:** Code B **Elevation:** 614.900 M
27A	869/R/A/W/T/ Concrete	171408.05 N 0782634.52 E	Pushback facing North or South on TWY H, Pull abeam stand 27A and then engine start-up approved **Suitability:** Code B **Elevation:** 614.800 M
27B	869/R/A/W/T/ Concrete	171407.03 N 0782634.54 E	Pushback facing North or South on TWY H, Pull abeam stand 27B and then engine start-up approved **Suitability:** Code B **Elevation:** 614.540 M
28	869/R/A/W/T/ Concrete	171408.03 N 0782632.92 E	Pushback facing East or West on TWY C, Pull abeam stand 28 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation (m):** 614.420 M
29	869/R/A/W/T/ Concrete	171408.01 N 0782631.54 E	Pushback facing East or West on TWY C, Pull abeam stand 29 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation (m):** 614.000
30	869/R/A/W/T/ Concrete	171408.00 N 0782630.18 E	Pushback facing East or West on TWY C, Pull abeam stand 30 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 613.650 M
31	869/R/A/W/T/ Concrete	171407.98 N 0782628.80 E	Pushback facing East or West on TWY C, Pull abeam stand 31 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 613.310 M
32	869/R/A/W/T/ Concrete	171407.96 N 0782627.01 E	Pushback facing East or West on TWY C, Pull abeam stand 32 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 612.750 M
33	869/R/A/W/T/ Concrete	171407.94 N 0782625.64 E	Pushback facing East or West on TWY C, Pull abeam stand 33 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation:** 612.350 M

34	869/R/A/W/T/ Concrete	171407.47 N 0782624.27 E	Pushback facing East or West on TWY C, Pull abeam stand 34 and then engine startup approved **Suitability:** Code C and Q 400 **Elevation (m):** 611.800
35	869/R/A/W/T/ Concrete	171407.45N 782622.44E	Pushback facing East or West on TWY C, Pull abeam stand 35 and then engine startup approved **Suitability:** Code C **Elevation:** 611.240 M
36	869/R/A/W/T/ Concrete	171407.43N 782621.07E	Pushback facing East or West on TWY C, Pull abeam stand 36 and then engine startup approved **Suitability:** Code C **Elevation:** 610.890 M
37	869/R/A/W/T/ Concrete	171407.41N 782619.70E	Pushback facing East or West on TWY C, Pull abeam stand 37 and then engine startup approved **Suitability:** Code C **Elevation:** 610.460 M
38	869/R/A/W/T/ Concrete	171407.39N 782618.33E	Pushback facing East or West on TWY C, Pull abeam stand 38 and then engine startup approved **Suitability:** Code C **Elevation:** 610.430 M
39	869/R/A/W/T/ Concrete	171405.65N 0782607.69E	Pushback on TXL K1 facing North or South, pull abeam stand 39 then engine start-up approved. **Suitability:** Code C **Elevation:** 607.760 M
40	869/R/A/W/T/ Concrete	171403.77N 0782608.63E	Pushback facing East or West on TWY E1, Pull abeam stand 40 and then engine startup approved **Suitability:** Code C **Elevation:** 607.760 M
41	869/R/A/W/T/ Concrete	171403.75N 0782607.16E	Pushback facing East or West on TWY E1, Pull abeam stand 41 and then engine startup approved **Suitability:** Code C **Elevation:** 607.660 M
42	869/R/A/W/T/ Concrete	171403.73N 0782605.68E	Pushback facing East or West on TWY E1, Pull abeam stand 42 and then engine startup approved **Suitability**: Code C **Elevation:** 607.540 M
43	869/R/A/W/T/ Concrete	171403.71N 0782604.21E	Pushback facing East or West on TWY E1, Pull abeam stand 43 and then engine startup approve **Suitability:** Code C **Elevation:** 607.430 M
44	869/R/A/W/T/ Concrete	171403.69N 0782602.74E	Pushback facing East or West on TWY E1, Pull abeam stand 44 and then engine startup approved **Suitability:** Code C **Elevation:** 607.360 M
45	869/R/A/W/T/ Concrete	171403.67N 0782601.27E	Pushback facing East or West on TWY E1, Pull abeam stand 45 and then engine startup approved **Suitability:** Code C **Elevation:** 607.340 M
46	869/R/A/W/T/ Concrete	171403.65N 0782559.80E	Pushback facing East or West on TWY E1, Pull abeam stand 46 and then engine startup approved **Suitability:** Code C **Elevation:** 607.400 M
47	869/R/A/W/T/ Concrete	171403.68N 0782558.32E	Pushback facing East or West on TWY E1, Pull abeam stand 47 and then engine startup approved **Suitability:** Code C **Elevation:** 607.390 M
48	869/R/A/W/T/ Concrete	171403.61N 0782556.85E	Pushback facing East or West on TWY E1, Pull abeam stand 48 and then engine startup approved **Suitability:** Code C **Elevation:** 607.340 M
49	869/R/A/W/T/ Concrete	171403.58N 0782555.38E	Pushback facing East or West on TWY E1, Pull abeam stand 49 and then engine startup approved **Suitability:** Code C **Elevation:** 607.390 M
50	869/R/A/W/T/ Concrete	171403.55N 0782553.92E	Pushback facing East or West on TWY E, Pull abeam stand 50 and then engine startup approved **Suitability:** Code C **Elevation:** 607.350 M

51	935/R/A/W/T/ Concrete	171403.53N 0782552.45E	Pushback facing East or West on TWY E, Pull abeam stand 51 and then engine startup approved **Suitability:** Code C **Elevation:** 607.370 M
52	935/R/A/W/T/ Concrete	171403.96N 0782550.80E	Pushback facing East or West on TWY E, Pull abeam stand 52 and then engine startup approved **Suitability:** Code C **Elevation:** 607.500 M
53	935/R/A/W/T/ Concrete	171403.34N 0782549.19E	Pushback facing East or West on TWY E, Pull abeam stand 53 and then engine startup approved **Suitability:** Code C **Elevation:** 607.310 M
54	935/R/A/W/T/ Concrete	171404.05N 0782546.91E	Pushback facing East or West on TWY E, Pull abeam stand 54 and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 54, stand 54L and 54R not available. **Suitability:** Code E **Elevation:** 607.490 M
54 L	935/R/A/W/T/ Concrete	171403.29N 0782546.43E	Pushback facing East or West on TWY E, Pull abeam stand 54L and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 54L, stand 54 not available. **Suitability:** Code C **Elevation:** 607.370 M
54 R	1093/R/A/W/ T/ Concrete	171404.24N 0782547.79E	Pushback facing East or West on TWY E, Pull abeam stand 54R and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 54R, stand 54 not available. **Suitability:** Code C **Elevation:** 607.490 M
55	935/R/A/W/T/ Concrete	171404.01N 0782544.20E	Pushback facing East or West on TWY E, Pull abeam stand 55 and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 55, stand 55L and 55R not available. **Suitability:** Code F **Elevation:** 607.480 M
55 L	935/R/A/W/T/ Concrete	171403.24N 0782543.47E	Pushback facing East or West on TWY E, Pull abeam stand 55L and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 55L, stand 55 not available. **Suitability:** Code C **Elevation:** 607.330 M
55 R	1424/R/C/W/ T/ Concrete	171404.37N 0782544.82E	Pushback facing East or West on TWY E, Pull abeam stand 55R and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 55R, stand 55 not available. **Suitability:** Code C **Elevation:** 607.670 M
56	1201/R/A/W/T Concrete	171403.95N 0782541.23E	Pushback facing East or West on TWY E, Pull abeam stand 56 and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 56, stand 56L and 56R not available. **Suitability:** Code F **Elevation:** 607.600 M
56 L	935/R/A/W/T/ Concrete	171403.19N 0782540.49E	Pushback facing East or West on TWY E, Pull abeam stand 56L and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 56L, stand 56 not available. **Suitability:** Code C **Elevation:** 607.300 M
56 R	935/R/A/W/T/ Concrete	171404.33N 0782541.86E	Pushback facing East or West on TWY E, Pull abeam stand 56R and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 56R, stand 56 not available. **Suitability:** Code C **Elevation:** 607.610 M
57	935/R/A/W/T/ Concrete	171403.91N 0782538.34E	Pushback facing East or West on TWY E, Pull abeam stand 57 and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 57, stand 57L and 57R not available. **Suitability:** Code E **Elevation:** 607.510 M

57 L	935/R/A/W/T/ Concrete	171403.15N 0782537.64E	Pushback facing East or West on TWY E, Pull abeam stand 57L and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 57L, stand 57 not available. **Suitability:** Code C **Elevation:** 607.270 M
57 R	935/R/A/W/T/ Concrete	171404.13N 0782539.00E	Pushback facing East or West on TWY E, Pull abeam stand 57R and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 57R stand 57 not available. **Suitability:** Code C **Elevation:** 607.540 M
58	935/R/A/W/T/ Concrete	171403.87N 0782535.49E	Pushback facing East or West on TWY E, Pull abeam stand 58 and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 58, stand 58L and 58R not available. **Suitability:** Code E **Elevation:** 607.530 M
58 L	935/R/A/W/T/ Concrete	171403.11N 0782534.79E	Pushback facing East or West on TWY E, Pull abeam stand 58L and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 58L stand 58 not available. **Suitability:** Code C **Elevation:** 607.270 M
58 R	935/R/A/W/T/ Concrete	171403.14N 0782536.15E	Pushback facing East or West on TWY E, Pull abeam stand 58R and then engine startup approved. MARS aircraft parking system, when ACFT parked on stand 58R stand 58 not available. **Suitability:** Code C **Elevation:** 607.540 M
59	935/R/A/W/T/ Concrete	171403.89 N 782532.62 E	Pushback facing East or West on TWY E, Pull abeam stand 59 and then engine startup approved MARS aircraft parking system, when stand 59R or 59L occupied stand 59 not available. **Suitability:** Code E **Elevation:** 607.400 M
59R	935/R/A/W/T/ Concrete	171403.26N 0782533.28E	Pushback facing East or West on TWY E, Pull abeam stand 59R and then engine startup approved. MARS aircraft parking system, when stand 59 occupied stand 59R & 59L not available **Suitability:** Code C **Elevation:** 607.350 M
59L	935/R/A/W/T/ Concrete	171403.20 N 782531.87 E	Push back facing East or West and pull abeam stand 59L and then engine startup approved.MARS aircraft parking system, when stand 59 occupied stand 59R & 59L not available. **Suitability:** Code C **Elevation:** 607.340 M
60	935/R/A/W/T/ Concrete	171403.85 N 782529.78 E	Pushback facing East or West on TWY E, Pull abeam stand 60 and then engine startup approved. MARS aircraft parking system,when stand 60R or 60L occupied stand 60 not available. **Suitability:** Code E **Elevation:** 607.200 M
60R	935/R/A/W/T/ Concrete	171403.18 N 782530.44 E	Pushback facing East or West on TWY E, Pull abeam stand 60R and then engine startup approved. MARS aircraft parking system, when stand 60 occupied stand 60R & 60L not available. **Suitability:** Code C **Elevation:** 607.000 M
60L	935/R/A/W/T/ Concrete	171403.16 N 782529.03 E	Pushback facing East or West on TWY E, Pull abeam stand 60L and then engine startup approved. MARS aircraft parking system, when stand 60L occupied stand 60 not available. **Suitability**: Code C **Elevation:** 606.860 M
61	935/R/A/W/T/ Concrete	171403.65N 0782527.87E	Push back deep facing East on TWY E and pull forward till stand 60L or deep pushback facing West till stand 60L or pushback facing south on to TWY M till TDP and then engine startup approved. **Suitability**: Code E **Elevation:** 606.630 M

62	935/R/A/W/T/ Concrete	171404.36N 0782526.27E	Pushback deep straight up to TWY M1 till the push back limit line and then pull forward on TWY M, facing south till TDP south of stand 62 and then engine startup approved. **Suitability:** Code C **Elevation:** 606.460 M
63	935/R/A/W/T/ Concrete	171406.76N 0782527.23E	Pushback facing south on TWY M and pull forward abeam stand 63 and then engine startup approved. **Suitability:** Code E **Elevation:** 607.100 M
63L	935/R/A/W/T/ Concrete	171407.60N 0782526.19E	Pushback facing south on TWY M and pull forward abeam stand 63L and then engine startup approved. **Suitability:** Code C **Elevation:** 605.898 M
63R	935/R/A/W/T/ Concrete	171406.29N 0782526.17E	Pushback facing south on TWY M and pull forward abeam stand 63R and then engine startup approved. **Suitability:** Code C **Elevation:** 606.720 M
64	935/R/A/W/T/ Concrete	171409.89N 0782527.13E	Pushback facing South on TWY M and pull forward abeam Stand 64 and then engine startup approved. **Suitability:** Code E **Elevation:** 607.000 M
65	869/R/A/W/T Concrete	171411.90N 0782526.09E	Pushback facing South on TWY M and pull forward abeam Stand 64 and then engine start up approved. **Suitability:** Code C **Elevation:** 607.100 M
66	869/R/A/W/T Concrete	171411.60N 0782528.96E	Pushback facing Northwest on TWY M2 and pull forward abeam Stand 66 and then engine startup approved. **Suitability:** Code C **Elevation:** 606.900 M
67	869/R/A/W/T Concrete	171411.08N 0782530.22E	Pushback facing Northwest on TWY M2 and pull abeam Stand 67 and then engine startup approved. **Suitability:** Code C **Elevation:** 606.900 M
68	869/R/A/W/T Concrete	171410.55N 0782531.48E	Pushback facing Northwest on TWY M2 and pull abeam Stand 67 and then engine startup approved. **Suitability:** Code C **Elevation:** 606.900 M
69	869/R/A/W/T Concrete	171409.65N 0782532.82E	Pushback facing Northwest on TWY M2 and pull abeam Stand 67 and then engine startup approved. **Suitability:** Code C **Elevation:** 606.205 M
70	906/R/B/W/T Concrete	171402.41N 0782618.31E	Push back facing North or South and pull/push abeam stand 71 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.510 M
71	906/R/B/W/T Concrete	171400.98N 0782618.33E	Push back facing North or South and pull/push abeam stand 71 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.370 M
72	906/R/B/W/T Concrete	171359.55N 0782618.35E	Push back facing North or South and pull/push abeam stand 71 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.160 M

73	906/R/B/W/T Concrete	171359.57N 0782619.84E	Push back facing North or South and pull/push abeam stand 74 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.490 M
74	906/R/B/W/T Concrete	171401.00N 0782619.82E	Push back facing North or South and pull/push abeam stand 74 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.860 M
75	906/R/B/W/T Concrete	171402.43N 0782619.80E	Push back facing North or South and pull/push abeam stand 74 and then engine startup approved. **Suitability:** Code C **Elevation:** 610.980 M
76	906/R/B/W/T Concrete	171402.49N 0782624.43E	Push back facing North or South and pull/push abeam stand 77 and then engine startup approved. **Suitability:** Code C **Elevation:** 612.370 M
77	906/R/B/W/T Concrete	171401.06N 0782624.45E	Push back facing North or South and pull/push abeam stand 77 and then engine startup approved. **Suitability:** Code C **Elevation:** 612.190 M
78	906/R/B/W/T Concrete	171359.63N 0782624.47E	Push back facing North or South and pull/push abeam stand 77 and then engine startup approved. **Suitability:** Code C **Elevation:** 611.900 M
79	906/R/B/W/T Concrete	171359.68N 0782628.63E	Push back facing North or South and pull/push abeam stand 80 and then engine startup approved. **Suitability:** Code C **Elevation:** 613.000 M
80	906/R/B/W/T Concrete	171401.11N 0782628.61E	Push back facing North or South and pull/push abeam stand 80 and then engine startup approved. **Suitability:** Code C **Elevation:** 613.400 M
81	906/R/B/W/T Concrete	171402.54N 0782628.59E	Push back facing North or South and pull/push abeam stand 80 and then engine startup approved. **Suitability:** Code C **Elevation:** 613.600 M
82	906/R/B/W/T Concrete	171402.60N 0782633.23E	Push back facing North or South and pull/push abeam stand 83 and then engine startup approved. **Suitability:** Code C **Elevation:** 614.940 M
83	906/R/B/W/T Concrete	171401.17N 0782633.25E	Push back facing North or South and pull/push abeam stand 83 and then engine startup approved. **Suitability:** Code C **Elevation:** 614.810 M
84	906/R/B/W/T Concrete	171359.74N 0782633.27E	Push back facing North or South and pull/push abeam stand 83 then engine startup approved. **Suitability:** Code C **Elevation:** 614.450 M
85	906/R/B/W/T Concrete	171359.76N 0782634.76E	Push back facing North or South and pull/push abeam stand 86 and then engine startup approved. **Suitability:** Code C **Elevation:** 614.860 M
86	906/R/B/W/T Concrete	171401.09N 0782634.74E	Push back facing North or South and pull/push abeam stand 86 and then engine startup approved. **Suitability:** Code C **Elevation:** 615.120 M
87	906/R/B/W/T Concrete	171402.63N 0782634.72E	Push back facing North or South and pull/push abeam stand 86 then engine startup approved. **Suitability:** Code C **Elevation:** 615.240 M

100	752/R/A/W/T Concrete	171356.51N 0782701.43E	Isolation stand
200	1201/R/B/W/T Concrete	171403.23N 0782514.56E	When ACFT parked on stand 200, stand 201 not available. **Suitability:** Code C Elevation: 602.000 M
201	935/R/A/W/T Concrete	171403.57N 0782513.76E	When ACFT parked on stand 201, stand 200 & 202 not available. **Suitability:** Code F **Elevation:** 602.100 M
202	935/R/A/W/T Concrete	171403.19N 0782512.97E	When ACFT parked on stand 202, stand 201 not available. **Suitability:** Code C Elevation: 602.000 M
203	114/R/B/W/T Rigid	171403.01N 0782511.49E	When ACFT parked on stand 203, stand 203L not available. **Suitability:** Code C Elevation: 601.900 M
203L	1201/R/B/W/T Concrete	171403.53N 0782510.54E	When ACFT parked on stand 203L, stand 204 & 203 not available. **Suitability:** Code F **Elevation:** 602.100 M
204	1201/R/B/W/T Concrete	171403.17N 0782509.86E	When ACFT parked on stand 204, stand 203L not available. **Suitability:** Code C Elevation: 602.000 M
NP 7	655/F/A/W/T Asphalt	171359.85 N 0782610.55 E	Aircraft Compatibility: Code “C” and Q 400 Power in Power out type stand. Elevation: 608.300 M
NP 8	655/F/A/W/T Asphalt	171359.80 N 0782606.93 E	Aircraft Compatibility: Code “C” and Q 400 Power in Power out type stand. Elevation: 607.800 M
NP 9	655/F/A/W/T Asphalt	171359.75 N 0782603.28 E	Aircraft Compatibility: Code “C” and Q 400 Power in Power out type stand. Elevation: 607.500 M
NP 10	655/F/A/W/T Asphalt	171359.71 N 0782559.66 E	Aircraft Compatibility: Code “C” and Q 400 Power in Power out type stand. Elevation: 607.500 M
NP 11	655/F/A/W/T Asphalt	171359.66 N 0782556.04 E	Aircraft Compatibility: Code “C” and Q 400 Power in Power out type stand. Elevation: 607.400 M

II. Pushback and Startup Procedures

1. GENERAL

i. Aircraft on all stands at RGI Airport, shall be pushed back from the stand towards the taxilane/taxiway centerline, taking into account the taxi route to be followed.

ii. SHAMSHABAD GROUND (frequency 121.850 MHz, or any alternate frequency) shall be contacted for pushback and startup permission only after the Pilot-In-Command has established that the aircraft is 'ready' for pushback/start up.

Note : For the purpose of clarity, 'ready' means all doors of the aircraft are closed, the Passenger Boarding Bridge (s) (PBB) / Step Ladder (s) are disconnected / withdrawn, the tug is connected to the aircraft and the ground engineer is in position and in contact with the Pilot-In-Command (PIC).

iii. On receiving the above mentioned information, ATC will allow pushback and startup of the aircraft depending on the traffic.

iv. ATC may deviate from the standard pushback procedures as stated for reasons such as traffic or work in progress.

v. Simultaneous push back from adjacent stands is not permitted.

Note : Adjacent stands shall include associated 'L'(left) and 'R' (right) stands except wherever specified otherwise.

vi. Simultaneous startup of aircraft engine(s) at adjacent stands may be allowed subject to the airline operator / GHA ensuring safety of personnel / aircraft / property.

vii. To approve the pushback and startup request from pilot, phraseology used by ATC maybe as given below :

"ATC : [*Call sign of Aircraft*] GROUND, RUNWAY--, PUSHBACK AND STARTUP APPROVED"

viii. When deviating from the standard pushback procedure due to any reason, ATC may use any other phraseology other than the one mentioned above.

ix. The pilot shall adhere strictly to SOP by aircraft manufacturer/ relevant operational manuals for starting up engine(s) at idle power or at breakaway locations.

x. At all times ground crew must strictly follow the safety instructions on hazard zones set out by aircraft manufacturer / relevant operational manuals.

xi. Deployment of wing walkers/wing guards by airline is mandatory prior to undertaking pushback from any stand at RGI Airport.

xii. On receipt of the clearance, the PIC shall read back the pushback clearance given by ATC, then co-ordinate with the ground crew for the pushback and start-up of the aircraft. The ground crew must ensure that the area in the front, behind and around the aircraft is clear of personnel, vehicles, equipment and other obstructions before commencing pushback and startup of the aircraft.

xiii. The PIC may start one engine, on idle power, before commencing pushback on the aircraft stand, in coordination with the ground crew.

xiv. Starting of second engine on idle power, before commencing pushback on the aircraft stand, shall be permitted after PIC confirms to ATC that the 2nd engine is being started with appropriate power unit and necessary ground precautions have been ensured.

xv. No cross-bleed startup by aircraft is permitted till the pushback and/or pull ahead procedure is complete and the Aircraft is aligned with the taxilane / taxiway centerline marking. Prior to starting the cross bleed, PIC shall seek confirmation from the ground crew for hazard free zone. At all times ground crew must follow the safety instructions on hazard zones set out by aircraft manufacturer/ relevant Operational manuals.

xvi. Ground crew of aircraft intending to do a cross bleed start must assess the clearance behind and in-front of aircraft and in case they feel that there is insufficient clearance, they shall request SMC for repositioning the aircraft to a position from where cross bleed start can take place. As a broad guideline a separation distance equivalent to 3 (three) aircraft length should be clear behind the aircraft so that there is no jet blast impact behind the aircraft due to cross bleed start.

xvii. Vehicle or movement of personnel is NOT permitted behind the aircraft once the anticollision beacons of the aircraft have been switched ON.

xviii. For aircraft stands without dedicated pushback lines, stand lead-in line may be used for pushback guidance.

xix. Tug Disconect Points have been provided, wherever necessary, for ease of identifying the point for releasing tug.

xx. Nomenclature for Tug Disconnect points begins with letter 'TDP' followed by a numeral.

xxi. Pilots shall adhere to the pushback and startup procedures and will use minimum breakaway power.

xxii. Simultaneous pushback (Code C Aircraft only) can be permitted in alternate stands. All aircraft shall pull forward till abeam lead in line of respective stand. Simultaneous pushback (Code E Aircraft only) can be permitted from stand 54 & 57, 55 & 58 stands i.e. separation of two Aircraft stands. All aircraft shall pull forward till abeam lead in line of respective stand.

xxiii. Aircraft on Stands 57L & 57 shall pushback till abeam lead in line of 58R for taxiing out from TWY E4 else pull forward till abeam respective stand.

xxiv. No Simultaneous pushback can be permitted on Stands 59R & 58. Simultaneous pushback can be permitted on stands 58R & 59R and 57L & 58L in case on 57L aircraft is pull forward till abeam 57L.

xxv. When B757 -200 parked on stand 59R, stand 58L and 59L shall be restricted for parking of Aircraft having max span 27.5 M or less (ATR-72-600).

xxvi. When aircraft is parked on L or R stand of 54, 55, 56, 57 & 58; centre stand of respective MARS configuration is not available or when aircraft parked on centre stand of 54, 55, 56,57 & 58; L or R stand of respective MARS configuration is not available.

xxvii. Aircraft on Stand 50, during pushback facing East, Aircraft shall pushback till abeam lead in line of stand 51 for taxiing out from TWY E3. In this case no simultaneous pushback permitted from stand 52. Aircraft on Stand 50, during pushback facing West, Aircraft shall pushback till abeam lead in line of stand 49 for taxiing out from TWY E3. In this case no simultaneous pushback permitted from stand 48.

xxviii. Aircraft on Stand 40 while pushback facing West, shall pull forward till abeam lead in line Stand 41 to maintain clearance from TWY K1. In this case no Simultaneous pushback can be permitted on Stand 42. Aircraft on Stand 40 while pushback facing East, shall pull forward till abeam lead in line Stand 40. In this case Simultaneous pushback can be permitted on Stand 42.

xxix. To utilize dynamic traffic pattern (from Stands 28 to 38 New Remote East Apron), opposite direction pushback is permitted subject to maintain two stand separation, where two simultaneous push back has to face opposite direction (nose facing away from each other in opposite direction) i.e. stand 31 facing west and stand 28 facing east in this scenario simultaneous pushback is permitted, in both the scenario aircraft shall pull forward abeam lead in line of their respective stand for engine start.

xxx. Stand 26B and 27A are capable for power in and power out type operation subject to stand 26A and 27B are kept vacant and all GSE equipment's are kept clear of ERA/ERL or advance staging area.

xxxi. Except stand 26A, 26B, 27A and 27B all stands 15 to 38 provided with Fuelhydrant system and Fuelling for stand 26A, 26B, 27A and 27B shall be done through bowsers. No VDGS system available for parking guidance and all

Aircrafts to park under the guidance of marshaller provided by respective Airlines/GHA.

xxxii. Intermediate holding positions are provided on taxiways/taxilanes as appropriate. However, the following restriction shall apply: When an aircraft holding on taxiway D, short of taxiway J, it will prohibit aircraft parking/pushback from stand 15 & 16 and vice versa. Similarly, when an aircraft holding on taxiway C, short of taxiway K, it will prohibit aircraft parking/pushback from stand 38 & 37 and vice versa.

xxxiii. Aircraft stand identification sign has been provided for all the parking stands with pole mounted stand identification sign.

xxxiv. Simultaneous pushback from alternate stands is permitted at remote east Apron subject to both Aircraft facing same direction, Aircraft facing opposite direction to maintain two stand separation.

xxxv. All stands 70 to 87 compatible to accommodate pushback facing North or South, however Aircrafts parked on north and south corner stands to ensure safe clearance available from Taxilane C1 and Taxiway B while pushing back.

xxxvi. Aircraft pushing back from stand 70 facing south shall pull forward abeam stand 71 for engine start and Aircraft pushing back from stand 72 facing north shall pull forward abeam stand 71 for engine start and follow further instructions from ATC.

xxxvii. Only one movement (Arrival or Departure) at a time shall be permitted from Taxilane H1, J1, J2, J3 and J.

xxxviii. VDGS available for parking guidance

xxxix. Only one pushback at a time shall be given from stand 61 to 63R.

xl. When B757 -200 parked on stand 59R, stand 58L and 59L shall be restricted for parking of Aircraft having max span 27.5 M or less (ATR-72-600).

xli. From Stands 200 - 204

a. All stands are remote parking with power in and push back.

b. Simultaneous pushback from cargo stand is not allowed.

c. The stand 201 & 203L is compatible to serve aircraft up to Code-F category.

d. Stand 200, 202, 203 & 204 are Code C compatible stands.

e. Stand 200 & 202 not available when Aircraft is parked on stand 201 and stand 203 and 204 not available when Aircraft parked on stand 203L.

f. Cargo stands 201 and 203L are provided with Fixed Electrical Ground Power (FEGP) units. Number and capacity of FEGPU - 2 each with the capacity of 90 KVA.

g. Hydrant fueling available.

h. VDGS and Stand Identification signs not available, surface marking provided.

i. Size of Cargo Apron is 190 X 281 M.

j. The aircraft shall pushback on TWY B facing east.

k. No aircraft movement is permitted from/to cargo stands during high power run up at TWY B.

2. PUSHBACK FROM STANDS

2.1 Power-in and power-out procedures for 05 stands NP7/NP8/NP9/ NP10/N P11:

2.1.1. Taxi Procedure:

1.1 Arrival RWY 09R

Vacate via L1/L2/L3/L4 - Cross RWY 27R - B3 -K - NP7

or

Vacate via L1/L2/L3/L4 - Cross RWY 27R - B3 - K - Taxilane E1 - NP8/NP9/NP10/NP11

1.2 Departure RWY 09R

NP11 - E3-B-B6 - RWY 27R - holding position RWY 09R - NP7/NP8/NP9/NP10 - power out facing west on Taxilane E1 - E3-B-B6 - RWY 27R - holding position RWY 09 R

or

NP7/NP8/NP9/NP10 - power out facing east on Taxilane E1 - K - RWY 27R - holding position RWY 09 R

1.3 Arrival RWY 27L

Vacate via R1, R2/R3/R4 - RWY 09L - B3 - NP7

or

Vacate via R2/R3/R4 - RWY 09L - B3 - Taxilane E1 - NP8/NP9/NP10/NP11

1.4 Departure RWY 27L

NP7/NP8/NP9/NP10 - power out facing west on Taxilane E1 - E3 – B -B2 - RWY 09L - holding position RWY 27L

or

NP7/NP8/NP9/NP10 - power out facing east on Taxilane E1- K- B- B2 - RWY 09L - holding position RWY 27L

NP11- E3- B- B2- RWY 09L - holding position RWY 27L

Note (NP7-NP11) :

i. Stands are remote parking with power-in and power-out facility.

ii. Nose wheel marking is provided on each stand facing west.

iii. Fuelling done through bowsers.

iv. VDGS not available.

v. Pilot shall taxi at speed of 15 KTs on Taxilane E1. Aircraft on stand NP7-NP11 shall power in and power out with minimum throttle.

vi. These stands are exclusively used for night halt arrangement between 1900 Hrs. IST to next day 0900 Hrs. IST for base parking and morning departures.

vii. Power in/out operations from stand NP7 to NP11 shall restrict pushback from stand 40 to 49, Conditional clearance for NP-7 to NP-11 subject to availability of taxilane E1 shall be implemented to ensure safe operations.

viii. Pilots shall obtain positive clearance from ground engineer for Engine start- up to commence power out maneuver and shall taxi with minimum throttle to avoid jet blast on adjacent stand user

Size of Main Apron - (Stand 40 to 62):1425 X 84M, Size of North west Apron (Stand 63L to 69):531X92 M.

Size of North East (Stand 03 to 14): 599 X 84 M

Size of Remote East Apron (stand 15 to 38) : 545 X 140 M

Size of South East Apron (stand 70 to 87) : 822 x 64 M

Size of Cargo Apron : 190 X 281M

III. Power in Power out provision for Code B Aircraft up to wing span of 20 m and length 21 m has been provided on stands 26B and 27A. Separate ground marking has been provided on these stands for Power in /Power out guidance. Details for Code B power in power out stands are as follows :

TABLE- 1A

Sl. No.	Aircraft Stand No.	Restriction	Remarks
1	26B	Stand 26A shall be kept vacant	Power Out with minimum thottle facing north and Join TWY H
2	27A	Stand 27B shall be kept vacant	Power Out with minimum throttle facing south and Join TWY H

IV. TAXIWAY:

Designator	Length/ Width (M)	PCR	Coordinates	Remarks
TWY A	82/25	566/F/A/W/T Asphalt	171353.76N 0782701.30E	Isolation Bay
TWY A1	427/25	825/F/A/W/T Asphalt	171348.40N 0782711.41E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY A2	225/25	825/F/A/W/T Asphalt	171348.22N 0782657.31E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY A3	225/25	825/F/A/W/T Asphalt	171347.94N 0782634.91E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY A8	225/25	825/F/A/W/T Asphalt	171346.99N 0782520.50E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY A9	225/25	1021/F/A/W/T Asphalt	171346.57N 0782448.05E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL for East Edge Blue(elevated) 60 M Max
TWY A10	282/25	825/F/A/W/T Asphalt	171346.53N 0782444.76E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL for West Edge Blue(elevated) 60 M Max
TWY B	2582.4/23	665/F/A/W/T Asphalt	17 13 56.90 N 0782553.54 E	1. Entire length of TWY B is suitable for Code E ACFT when Secondary RWY 09L-27R in use. 2. TWY B from west of TWY B4 to CARGO APRON suitable for Code F ACFT when Main RWY 09R-27L in use and rest of the portion from east of TWY B4 to TWY B2 suitable for Code-E Acft. C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL at TWY junctions Blue(elevated) 60 M Max
TWY B1	165/23	862/F/A/W/T Asphalt	171357.22N 0782657.19E	MRO Apron Suitable for all aircraft up to Code F C/L: None Edge lights: AVBL Blue(elevated) 60 M Max
TWY B2	175/23	797/F/A/W/T Asphalt	171354.60N 0782637.27E	Suitable for all aircraft up to Code E C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max

TWY B3	175/25	895/F/B/W/T Asphalt	171357.19 N 0782613.05 E	Suitable for all aircraft up to Code E C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY B4	175/25	943/F/C/W/T Asphalt	171355.51N 0782553.79E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY B5	175/25	745/F/A/W/T Asphalt	171353.78 N 0782538.29 E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY B6	175/25	771/F/C/W/T Asphalt	171353.49N 0782520.41E	Suitable for all aircraft up to Code F C/L: None, Edge lights: AVBL for south Edge Blue(elevated) 60 M Max
TXL C	712.5/15.0	584/F/B/W/T Asphalt	171405.57 N 0782625.06 E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max
TXL C1	712.5/15.0	655/F/A/W/T Asphalt	171404.25N 0782625.08E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max
TXL D	779/18.6	522/F/C/W/T Asphalt	171411.67 N 0782623.26 E	Code-C, Suitable for all aircraft up to wing span 36M C/L: None Edge lights: AVBL for North Edge, Blue (elevated) 60M Max Shoulder:3.5M
TWY E	969.7/25	758/F/C/W/T Asphalt	171359.88 N 0782537.22 E	Suitable for all aircraft up to Code F C/L: None, Edge lights: AVBL for south Edge Blue(elevated) 60 M Max
TXL E1	569.7/18.6	655/F/A/W/T Asphalt	171401.17 N 0782601.87 E	Suitable for aircraft with max wing span up to 36M
TWY E3	97.0/25.0	797/F/C/W/T Asphalt	171358.43N 0782553.79E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY E4	97.0/25.0	745/F/A/W/T Asphalt	171358.20N 0782538.19E	Suitable for all aircraft up to Code F C/L: AVBL, Green(inset) 30 M Max Edge lights: AVBL Blue(elevated) 60 M Max
TXL H	190/18.6	534/F/B/W/T Asphalt	171408.09 N 0782637.08 E	Code C, Suitable for all aircraft up to wing span 36M C/L: None Edge lights: AVBL for East Edge, Blue(elevated) 60 M Max Shoulder:3.5
TXL H1	254.80/15.0	557/F/A/W/T Asphalt	171401.59N 0782637.17E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max Edge lights: AVBL for East Edge, Blue(elevated) 60 M Max Shoulder:3.5M

TWY K	544/25	655/F/A/W/T Asphalt	171404.28 N 0782613.00 E	Code C, Suitable for all aircraft up to max wing span 36M C/L: None Edge lights: AVBL, Blue(elevated) 60 M Max
TXL K1	412/18	708/F/C/W/T Asphalt	171406.38 N 0782610.16 E	Code C, Suitable for all aircraft up to max wing span 36M C/L: None Edge lights: AVBL, east Edge Blue(elevated) 60 M Max
TXL K2	621/15.0	655/F/A/W/T Asphalt	171413.27N 0782605.22E	Code C, Suitable for all aircraft up to max wing span 36M C/L: None Edge lights: None
TXL K3	622/15.0	708/F/B/W/T Asphalt	171414.57 N 0782604.76	Code C, Suitable for all aircraft up to max wing span 36M and max outer main wheel gear span up to 9.6M C/L: None Edge lights: AVBL, North Edge Blue (elevated) 60 M Max
TWY L1	438/15	581/F/B/W/T Asphalt	171347.46N 0782546.614E	1. RET 2. Suitable for all aircraft upto Code-C category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY L2	438/23	1021/F/A/W/ T Asphalt	171347.60N 0782557.60E	1. RET 2. Suitable for all aircraft upto Code-E category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY L3	499/25	1021/F/B/W/ T Asphalt	171347.58N 0782606.83E	1. RET 2. Suitable for all aircraft upto Code F category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY L4	499/25	745/F/A/W/T Asphalt	171347.81N 0782624.67E	1. RET 2. Suitable for all aircraft upto Code F category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY R1	438/15	708/F/B/W/T Asphalt	171347.77N 0782610.55E	1. RET 2. Suitable for all aircraft upto Code-C category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY R2	438/23	825/F/A/W/T Asphalt	171347.63N 0782559.56E	1. RET 2. Suitable for all aircraft upto Code-E category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY R3	482/25	745/F/A/W/T Asphalt	171347.30N 0782544.68E	1. RET 2. Suitable for all aircraft upto Code F category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TWY R4	482/25	745/F/A/W/T Asphalt	171347.10N 0782529.43E	1. RET 2. Suitable for all aircraft upto Code F category 3. C/L: AVBL, Inset 15M Max Edge lights: AVBL Blue(elevated) 60 M Max
TXL M	579.51/23	825/F/A/W/T Asphalt	171405.91N 0782523.33E	Suitable for all aircraft up to Code E C/L: None Edge lights: AVBL ,Blue(elevated) 60 M Max

TXL M1	579.51/23	825/F/A/W/T Asphalt	171405.88N 0782520.90E	Suitable for all aircraft up to Code E C/L: None Edge lights: AVBL , Blue(elevated) 60 M Max
TXL M2	461.06/15	655/F/A/W/T Asphalt	171414.10N 0782528.57E	Suitable for all aircraft upto Code- C category C/L: NA Edge lights: AVBL, Blue(elevated) 60 M Max
TXL M3	370.57/15	655/F/A/W/T Asphalt	171415.38N 0782527.03E	Suitable for all aircraft upto Code- C category C/L: NA Edge lights: AVBL, Blue(elevated) 60 M Max
TXL J	280.5/15	708/F/B/W/T Asphalt	171408.35N 0782615.76E	Code C, Suitable for all aircraft up to max wing span 36M C/L: None Edge lights: AVBL, east Edge Blue(elevated) 60 M Max
TXL J1	254.5/15.0	708/F/B/W/T Asphalt	171401.51N 0782630.93E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max
TXL J2	254.8/15.0	655/F/A/W/T Asphalt	17 14 00.39 N 078 26 22.13 E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max
TXL J3	254.8/15.0	655/F/A/W/T Asphalt	17 14 01.321 N 078 26 15.89 E	Code C, Suitable for all aircraft up to wing span 36M C/L: AVBL Green (inset) 30M Max Edge light: AVBL (west shoulder) Blue(elevated)60M Max

NOTE:

1. When Secondary RWY 09L-27R is active and used for landing and take-off entire length of taxiway B will be suitable for operations of Aircrafts having maximum wing span of 65M or less.(i.e. upto Code E Aircrafts) .
2. When Secondary RWY 09L-27R is used as parallel Taxiway then portion of TWY B between TWY B4 to TWY B6 till west side upto cargo Apron end is suitable for operations of aircraft having maximum wing span of 80M or less (i.e. upto code F aircrafts) and remaining portion between east of TWY B4 to TWY B2 will remain restricted for aircraft having maximum wing span 65M (i.e. Code E Aircrafts).
3. Intermediate holding position provided west of stand 60L on TWY E for holding the Aircraft clear off TWY M, M1 & E Junction to accommodate efficient traffic flow.
4. Dual Taxilanes C & C1 is commissioned at Remote Apron to cater the taxiing requirements of aircraft for 26 remote east apron stands and 18 south east apron stands.
5. Taxiway C centerline shifted 3.5 m towards north to accommodate new parallel taxilane C1. East of realigned Taxilane C is connected to Taxilane H and west of Taxilane C is connected to TWY K. Taxilane C is connected to stands 28 to 38 towards north and provide access to taxilane J1, J2 & J3 towards south via taxilane C1.
6. Taxilane C1 commissioned parallel to TWY C towards south at a distance of 40.5M from centerline of Taxilane ‘C’. Eastern end of taxilane C1 is connected to TWY H - H1 and western end of Taxilane C1 is connected to TWY K. The new Taxilane C1 will provide access to stand 28 to 38 via Taxilane C towards north and connect stands 70 to 87 via taxilane J1, J2 & J3 towards south.
7. Simultaneous taxiing on dual code C taxilane C & C1 is permitted subject to maintaining adequate separation from Aircraft pushing from stand 28 to 38.
8. When deviating from the standard pushback procedure due to any reason, ATC may use appropriate phraseology to maintain safe clearance from other traffic using aerodrome.
9. Vehicle or movement of personnel is NOT permitted behind the aircraft once the anticollision beacons of the aircraft have been switched ON.
10. For aircraft stands without dedicated pushback lines, stand lead-in line may be used for pushback guidance.
11. Aircraft holding at Intermediate holding position at TWY E facing west shall block pushback & access to stand 60L.
12. Taxiway B3 is re-configured and made compliant to accommodate code E operations till short of stand NP7 and TWY K junction towards south.
13. Code-E Aircraft can access taxiway 'B' through TWY B3 for through taxi route to TWY 'B2' and vice versa.
14. TWY B2 Commissioned ,CL / EDGE / RHP / Enhanced TWY CL at TWY B2 / Intermediate Holding PSN Markings are provided
15. Both sides of taxiway B2 with a pair of elevated guard lights flashing alternate yellow facing the taxiway is provided.
16. TWY A1, A2, A3, A8, A9, A10, CL LGT Available only at visibility conditions of RVR 1000M or less.
17. Rapid exit TWY CL LGT corresponding to the RWY in use Available during the time of operations of aeronautical ground lighting.
18. VFR training FLTs north of Nadirgul AD in a radius of 2NM. Lower limit: GND Upper limit: 2600FT AMSL.
19. Storm Water Drain on either side of RWY 27L/09R DIST from CL 105M, Shape trapezoid, Width at the top 10M, Width at the bottom 2M to 3M, Depth 2M to 3M.
20. TWY edge lights with spacing of 60m available on RWY 09L/27R when used exclusively for taxiing purpose.
21. Rapid Exit Taxiway L1 & R1 is suitable for All Aircraft up to Code C category or Aircraft having max wing span up to 36M.

22. Rapid Exit Taxiway L2 & R2 is suitable for All Aircraft up to Code E category or Aircraft having max wing span up to 65M.
23. Intermediate holding position marking to hold short TXL E1 provided on TWY K1 and to hold short of TWY B is provided on TWY K.
24. Location of RET L1 for RWY 09R is east of TWY A8 at a distance of 1450 M from RWY 09R threshold.
25. Location of RET L2 for RWY 09R is east of TWY A8 at a distance of 1775 M from RWY 09R threshold.
26. Location of RET R1 for RWY 27L is west of TWY A3 at a distance of 1425 M from RWY 27L threshold.
27. Location of RET R2 for RWY 27L is west of TWY A3 at a distance of 1750 M from RWY 27L threshold.
28. Aircraft on all stands at RGI Airport, shall be pushed back from the stand towards the taxilane/taxiway centreline, taking into account the taxi route to be followed.
29. Code C compliant Dual Taxilanes K2 & K3 is commissioned at North-East Apron to cater the taxiing requirements of aircraft for stands 3 to 12.
30. Taxilane K3 and K2 are parallel TXL and located North of stand 3 to 12 and connected to TXL K1, TWY K and TXL J towards east, Taxilane K2 and K3 centreline are separated at a distance of 40.5M between stand 5 to 12.
31. Taxilane K2 and K3 capable for parallel operations between stand 5 to 12, however no parallel operations shall be conducted beyond lead in line of stand 5 towards west and only one movement either arrival or departure permitted abeam stand 4 and 3.
32. Taxilane K3 preferably used for arrival movement and provides access to taxiing in aircrafts to stands 3 to 12 and Taxilane K2 preferably used for departure movement and provides access to taxiing out aircrafts from stands 3 to 12.
33. Simultaneous taxiing of code C aircraft on Taxilanes K2 & K3 are permitted subject to maintaining adequate separation from aircraft pushing from stands 5 to 12.
34. Pilots to taxi with minimum speed to avoid jet blast on service road north of Taxilane and maintain heightened attention at service road crossing at junction of TXL K1, J and TWY K. with TXL K2 and K3.
35. Extended portion of Code E compliant dual Taxilane M- M1 and Code C Compliant Dual Taxilanes M2 & M3 is commissioned at North-West Apron to cater the taxiing requirements of aircraft for stands 62 to 69.
36. Taxilane M - M1 and M2 - M3 are parallel TXL and located west of stand 62 to 65 and North of stand 66 to 69 respectively. South end of TXL M-M1 connected to TWY E and North End to TXL M2 -M3, similarly west end of TXL M2- M3 is connected to TXL M -M1 and east end merging abeam stand 67. Taxilane MM1 centreline is separated at 72.5M and M2 - M3 centreline is separated at a distance of 40.5M upto stand 66.
37. Taxilane M- M1(code E) and M2 -M3(Code C) capable for simultaneous parallel operations of compatible Aircraft. TXL M2-M3 from junction of TXL M&M1 upto lead in line of stand 66 is compliant for simultaneous parallel operations and portion beyond stand 66 towards southeast end upto stand 69 not suitable for simultaneous parallel operation. Only one movement either arrival or departure permitted from stand 67 to 69 on Taxilane M2.
38. Taxilane M1 & M3 preferably used for arrival movement and provides access to taxiing in aircrafts from stand 62 upto stand 67, beyond lead in line of stand 67 TXL M3 merges with TXL M2 and provides access to stands 68 and 69. Taxilane M& M2 preferably used for departure movement and provides access to taxiing out aircrafts from stands 69 to 62.
39. Simultaneous taxiing aircraft on Taxilanes M-M1 (code E) and M2 - M3(code C) respectively are permitted subject to maintaining adequate separation from aircraft pushing back from stand 62 to 66.
40. Pilots to taxi with minimum speed to avoid jet blast on service road of Taxilane and maintain heightened attention at service road crossing at junction of TWY M & M1.

41. ADS Commissioning with following Details

FAC : ADS-B

FREQ : 1090MHz

COORD : 171457N 0782628E

SITE ELEV : 616.50 M AMSL

MAX Range upto 250M.

VOHS AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart(16 April 2026)
2. Aerodrome Chart- Hot Spot(16 April 2026)
3. Aircraft Parking/Docking Chart (Passenger Terminal Apron)(16 April 2026)
4. Aircraft Parking/Docking Chart EAST Remote Apron(16 April 2026)
5. Aircraft Parking/Docking Chart Cargo Apron(16 April 2026)
6. Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 09R (12 June 2025)
7. Aerodrome Obstacle Chart Type A(Operating Limitations) RWY 27L(12 June 2025)
8. Aerodrome Obstacle Chart Type A(Operating Limitations) RWY 09L(12 June 2025)
9. Aerodrome Obstacle Chart Type A (Operating Limitations) RWY 27R(12 June 2025)
10. Aerodrome Obstacle Chart - Type B(12 June 2025)
11. Precision Approach Terrain Chart RWY 09R (12 June 2025)
12. Precision Approach Terrain Chart RWY 27L(12 June 2025)
13. Precision Approach Terrain Chart RWY 09L(12 June 2025)
14. Precision Approach Terrain Chart RWY 27R (12 June 2025)
15. TORA from Different Runway Intersections – Departure RWY 09R
16. TORA from Different Runway Intersections – Departure RWY 27L
17. Locations of Different Exit Taxiways from Threshold – Arrival 09R
18. Locations of Different Exit Taxiways from Threshold – Arrival 27L
19. Isolation Bay Chart (16 April 2026)
20. Locations of Different Exit Taxiways from Threshold – Arrival 09L
21. Locations of Different Exit Taxiways from Threshold – Arrival 27R
22. TORA from Different Runway Intersections – Departure RWY 09L (01 OCT 2026)
23. TORA from Different Runway Intersections – Departure RWY 27R (01 OCT 2026)
24. ILS Critical and Sensitive Area - RWY 09R - 27L(16 April 2026)
25. ILS Critical and Sensitive Area - RWY 09L - 27R(16 April 2026)
26. VOR Procedure RWY 09L (17 April 2025)
27. VOR Procedure RWY 09R (17 April 2025)
28. VOR Procedure RWY 27L (17 April 2025)
29. VOR Procedure RWY 27R (17 April 2025)
30. RNP Y Procedure RWY 09L
31. Coding Table - RNP Y Procedure RWY 09L
32. ILS Procedure RWY 09L
33. Coding Table - ILS Procedure RWY 09L
34. RNP Y Procedure RWY 27R
35. Coding Table - RNP Y Procedure RWY 27R
36. RNP Y Procedure RWY 27L
37. Coding Table - RNP Y Procedure RWY 27L

38. ILS or LOC Procedure RWY 09R (17 April 2025)
39. RNP Y Procedure RWY 09R (17 April 2025)
40. Coding Table - RNP Y Procedure RWY 09R (17 April 2025)
41. LVP Taxi Routing Easterly Flow RWY 09R CAT -III (ARRIVAL AND DEPARTURE)(16 April 2026)

VOHS AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	**2**	**3**